

Unit : 3

Transmission Media

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COURSE OUTCOME (CO3)

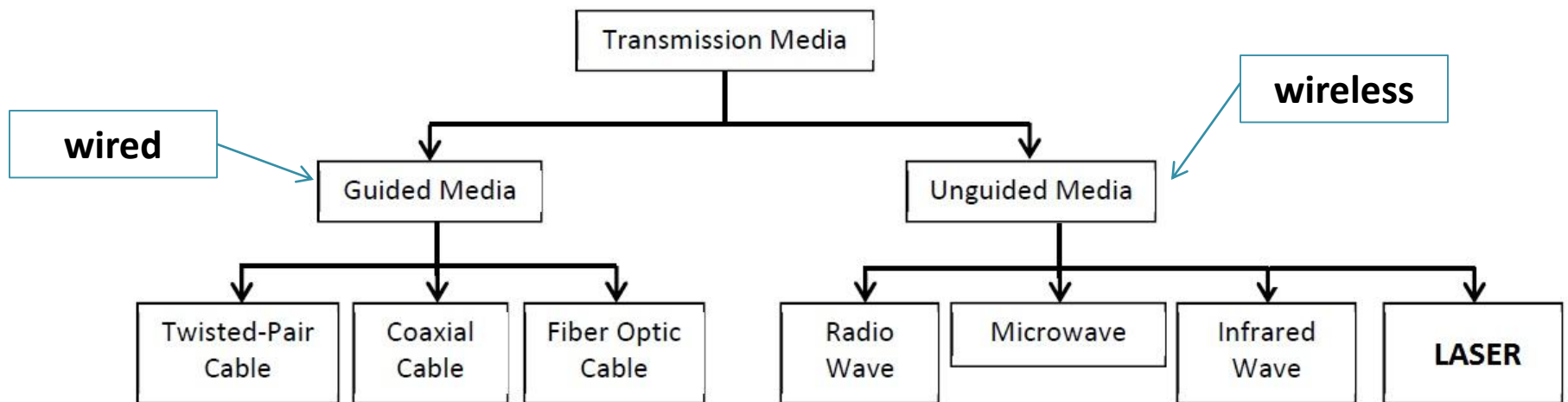
- After the completion of this unit, students will be able to.... *Illustrate use of transmission media used in computer network.*

LEARNING OUTCOME

- After the completion of this session, students will be able to.... *list out guided and unguided transmission media*

3.1 Types of Transmission Media

- Transmission media is the way through which the data, information or messages can flow between the nodes/workstations/computers. **OR**
- *A transmission media can be defined as anything that can carry information from a source to a destination. It should be air, water, cable etc.*
- Transmission media can be classified in to two broad categories: Guided media and Unguided media.



LEARNING OUTCOME

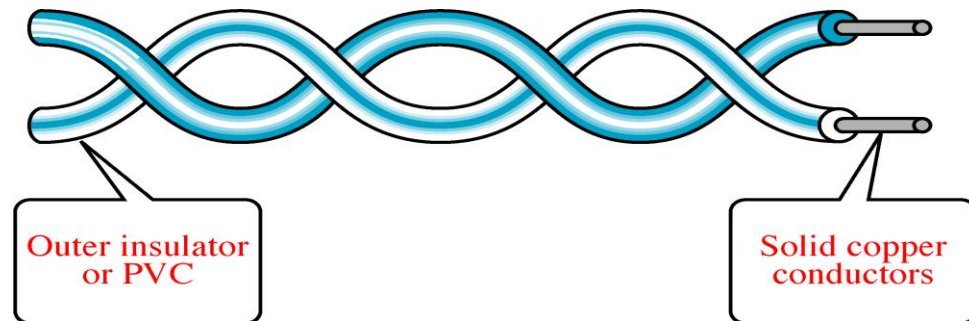
- After the completion of this session, students will be able to.... *select appropriate guided transmission media for given network.*

3.2 Guided Media (Physical media)

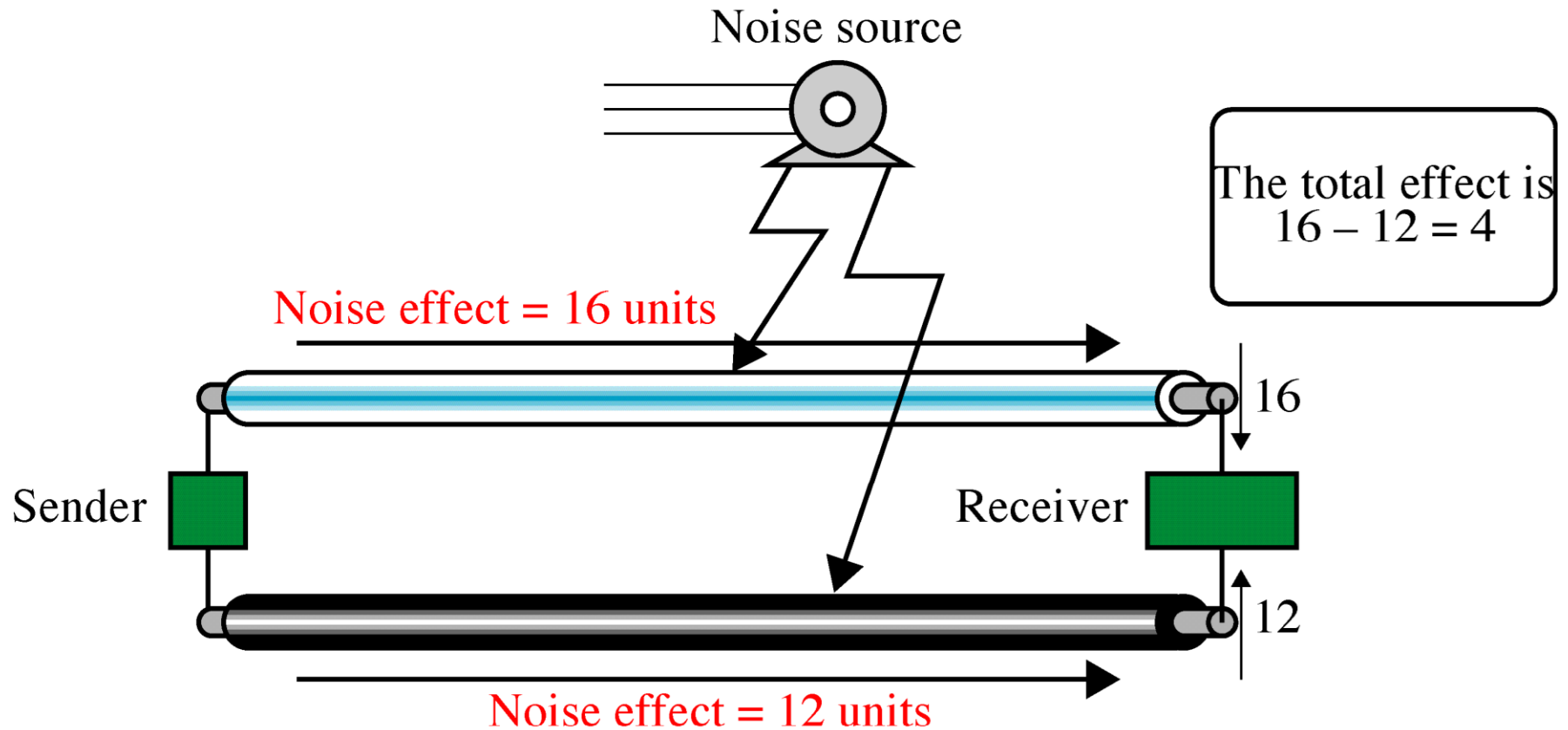
- Transmission Media uses a "cabling" system that guides the data signals along a specific path.
- It provides a conduit from one device to another
- It is also known as '**bounded media**' or '**wired communication**'.
- Examples of guided media are: twisted-pair cable, coaxial cable, and fiber-optic cable.

❖ Twisted pair cable

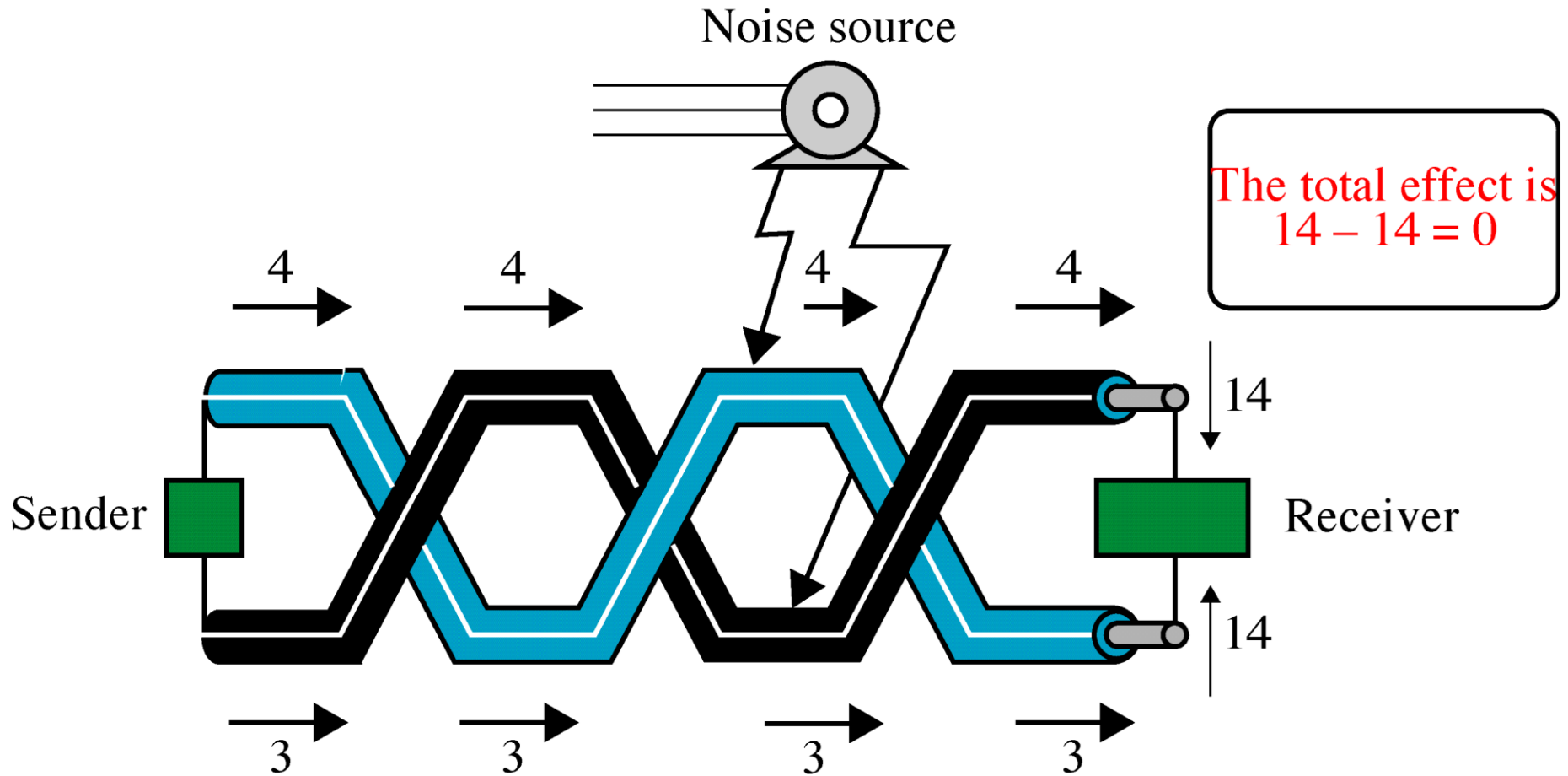
- Twisted pair cable comes into two forms → *unshielded and shielded twisted pair*.
- Both of these types uses metallic (copper) conductor that accept and transport signals in the form of electric current.



Effect of Noise on Parallel Lines (Extra)

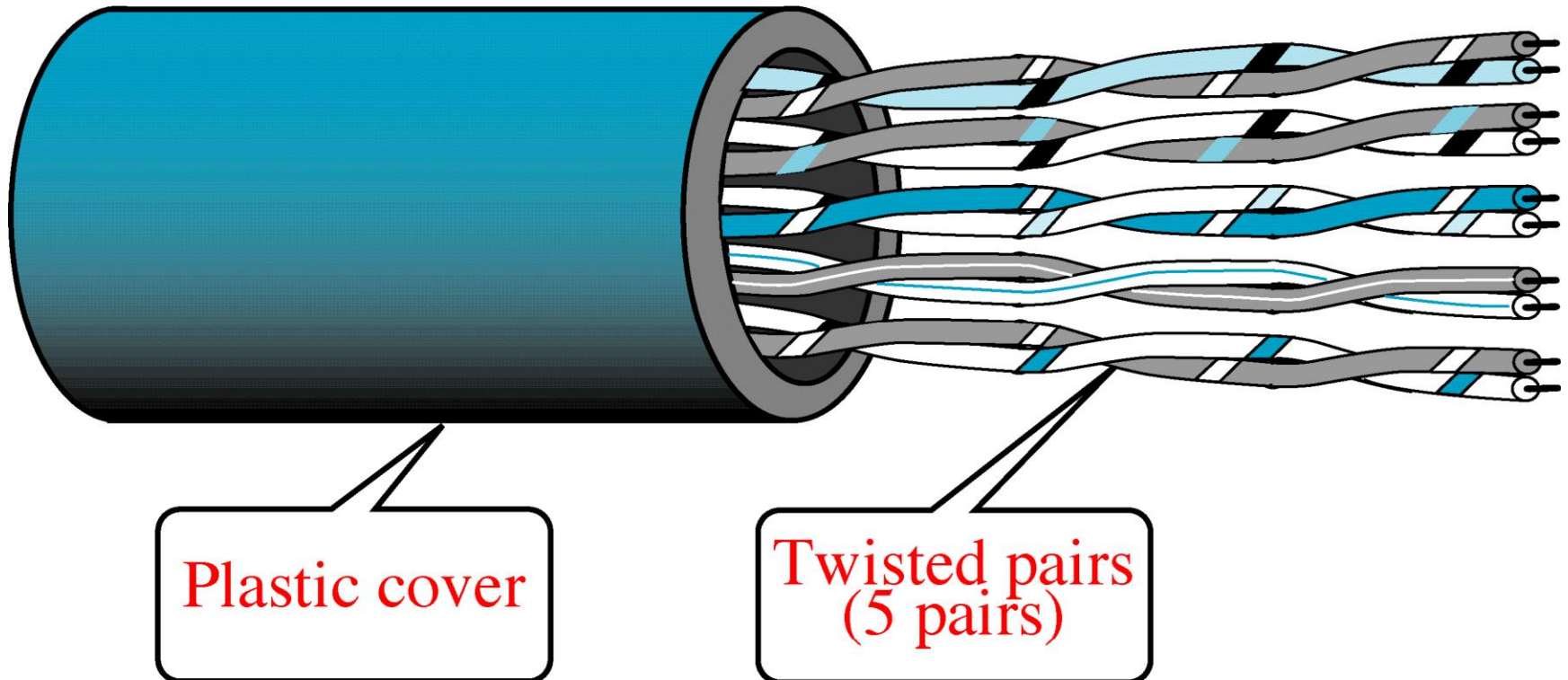


Noise on Twisted-Pair Lines (Extra)



❑ Unshielded twisted pair cable (UTP)

- It is most common type of medium used now a days.
- It is mostly used in telephone system.
- It's frequency range is suitable for both data and voice.
- Frequency range is 100 hz to 5mhz.



➡ **Advantages of UTP**

- Low cost
- Easy to use
- Cheap and flexible
- Easy to install
- Used in many LAN technology like Ethernet and token ring

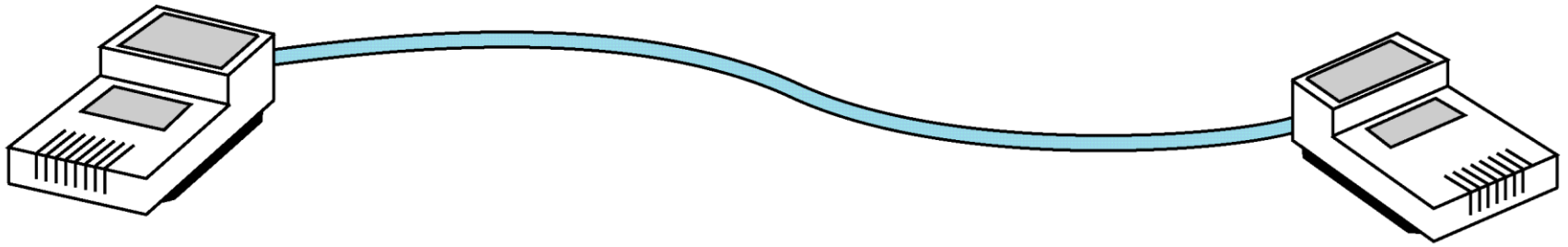
➡ **Disadvantages of UTP**

- It can be affected by EMI (electromagnetic interference) which can create noise and damage the signal.
- Electronics Industries Association (EIA) has categorized UTP by quality are:
- Category 1
 - Used in telephone system
 - Low speed data communication
- Category 2
 - Suitable for voice and data up to 4 mbps

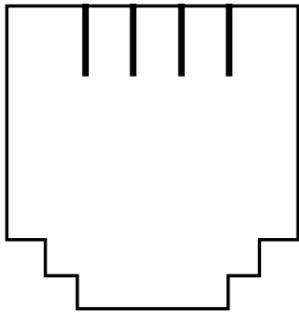
- Category 3
 - It can be used for data transmission up to 10mbps.
 - It is now used in standard telephone system.
- Category 4
 - It can be used for data transmission up to 16mbps.
- Category 5
 - It can be used for data transmission up to 100mbps.

◆ **UTP Connectors**

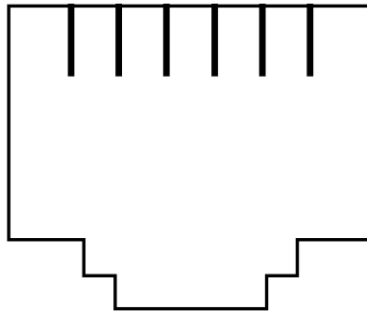
- UTP is most commonly connected to network devices via a type of snap in plug like that used with telephone jacks.
- Connectors are either male (the plug) or female (the receptive).
- Each wire in a cable is attached to one conductor (or pin) in the connector.
- The most frequently use of these plugs is an RJ45 (Registered Jack), connector with 8 conductor or RJ11 connector with 4 conductor (see figure).



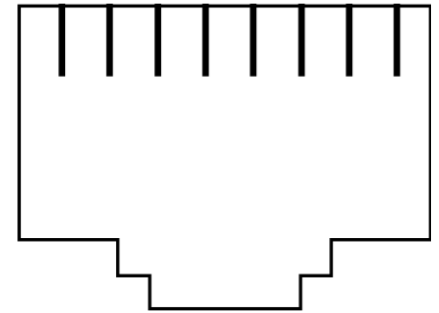
4-conductor



6-conductor



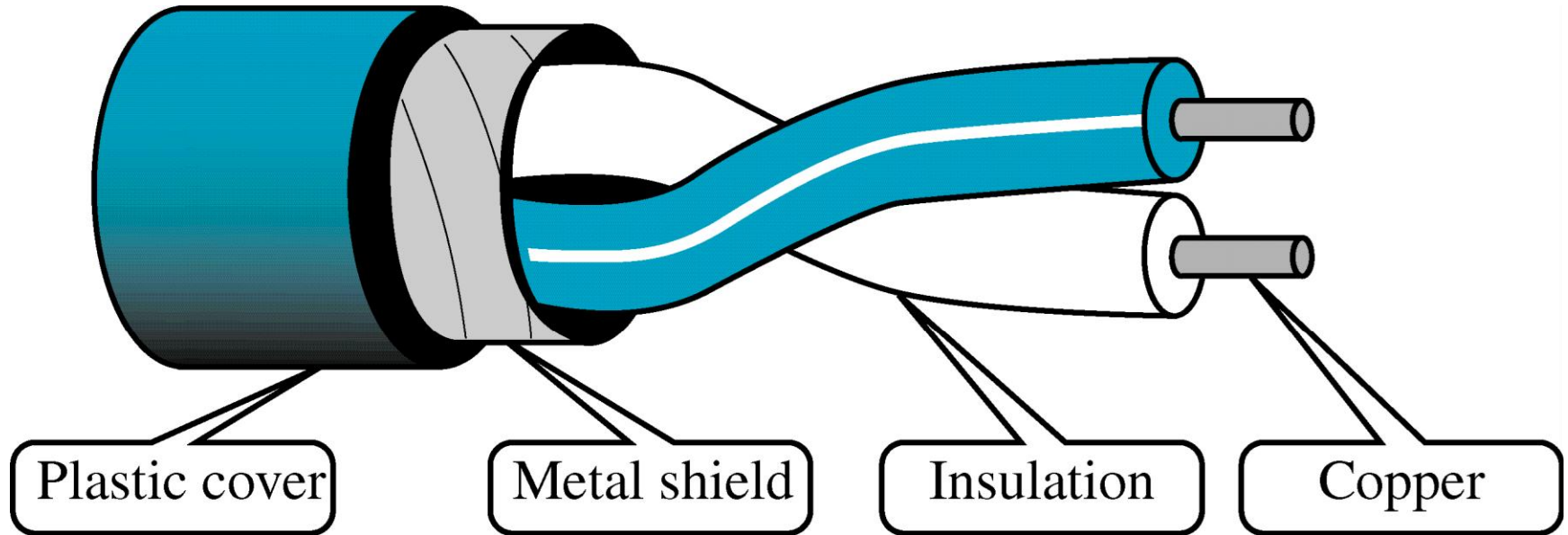
8-conductor



RJ11 connector with 4 conductor,
used in telephone system

RJ45 connector with 8 conductor,
used in networking

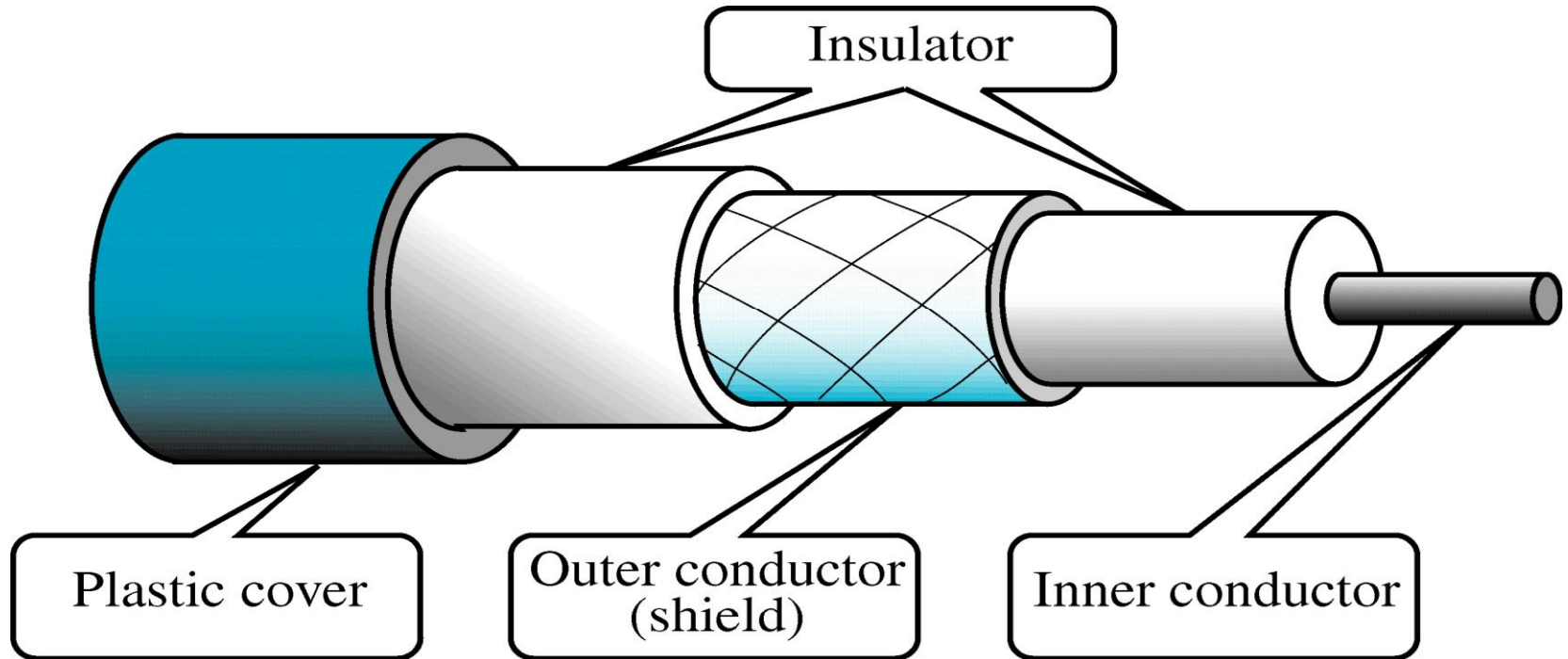
❑ Shielded twisted pair cable (STP)



- STP cable has a metal foil or braided-mesh covering that **encases each pair of insulated conductors**, which are a higher quality and more protective jacket than UTP has.
- This gives STP excellent insulation to protect the transmitted data from outside interference (Noise).
- It is less susceptible to electrical interference and supports higher transmission rates over longer distance than UTP.

- It has also a drawback that it is more expensive than UTP.
- ◆ **STP connector**
- STP uses the same connectors as UTP but the shield must be connected to ground.

❑ Coaxial cable



- This cable has better shielding than twisted pair, so it can span longer distance at higher speed.
- As we can see in the figure, it has central core conductor of solid or standard wire (usually copper) enclosed in an insulating sheath, which is in turn encased in an outer conductor of metal foil, braid or combination of two.

- The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit.

↪ **Advantages of coaxial cable**

- It is most widely used in network cable
 - It is relatively inexpensive, light, flexible and easy to work with.
 - Installation is very easy.
 - It is good choice for longer distance and for reliably supporting higher data rates.
- There are two types of coaxial cable
 1. Thin (Thinnet)
 2. Thick (Thicknet)

◆ **Thinnet coaxial cable**

- It is a flexible coaxial cable about 0.25 inches thick.
- Easy to install and work with it.

- it can be used in almost any type of network installation almost in BUS topology.
- can carry a signal up to approximately 185 meters (about 607ft) before the signal starts to suffer from attenuation.
- It is referred to the RG-58 (radio grade) family and has 50 - ohm impedance.
- Commonly used for digital transmission.

◆ Thicknet coaxial cable

- Thicknet is relatively rigid coaxial cable about 0.5 inches in diameter (Almost Double of thin net coaxial cable).
- Thick net can carry signal for 500 meters [about 1640 ft].
- Because of thicknet's ability to support data transfer over longer distance it is sometimes used as backbone to connect several smaller thinnet based networks.

- A device called a transceiver connects the thinnet coaxial to the larger thicknet coaxial cable.
- RG-8, RG-9 and RG-11 used for thicknet coaxial cable.

◆ Coaxial cable connectors

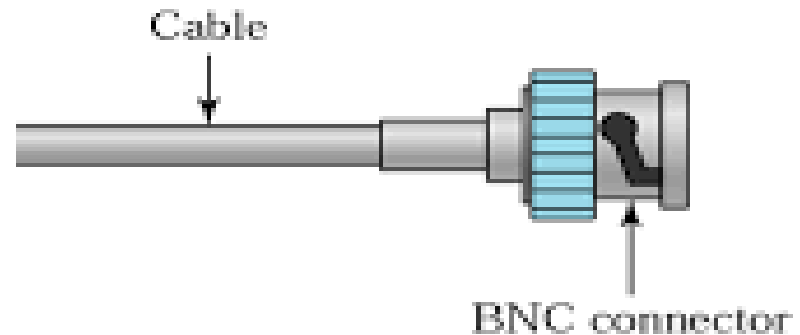
- There are four types of coaxial cable connectors
 1. Barrel connectors
 2. BNC (bayonet network connector)
 3. T-connectors
 4. Terminators

↪ Barrel connectors



- It is the most common connector.
- Generally this connector is used to connect BNC coaxial cable to increase the length of network.

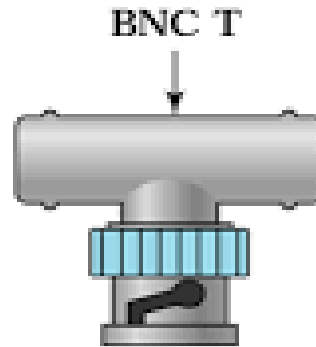
➔ BNC connectors



- After the Barrel connector, the most popular is the BNC, which pushes on and locks in to place with a half turn as shown in fig.
- Generally, a cable terminates in a male connector that plugs or screws on to a corresponding female connector attached to the device.
- Generally **used in cable TV and VCR hook-ups.**

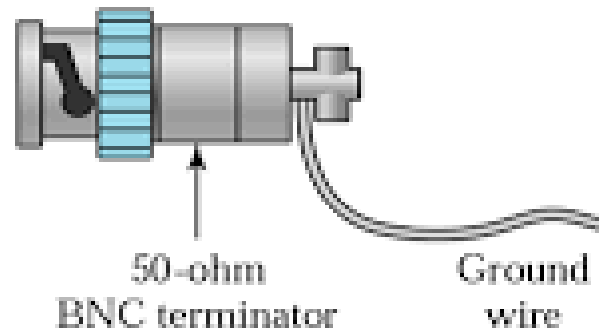
- These connectors are used to connect RG-58 or RG-62 with the NIC.
- And these are generally used in BUS and STAR topology.

➡ **T - connectors**

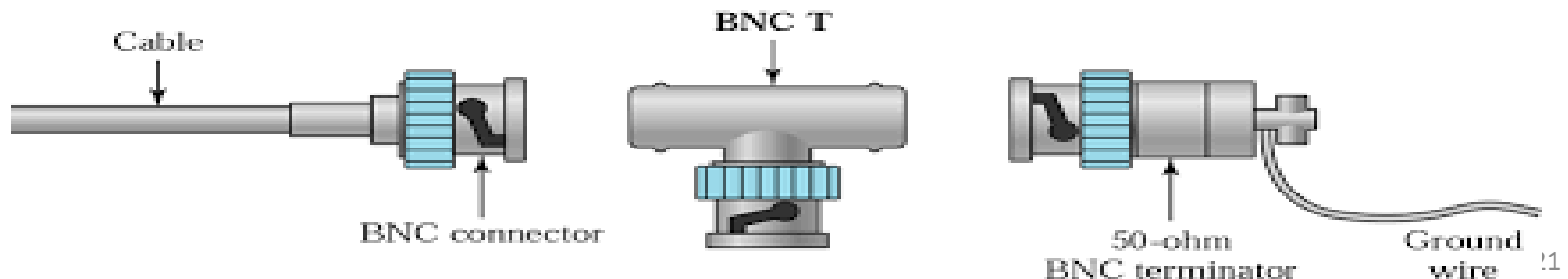


- T-Connectors (used in thin Ethernet) allow a secondary cable or cables to branch off from a main line.
- It is also **used to extend the network or increase the number of node at any point in the network.**

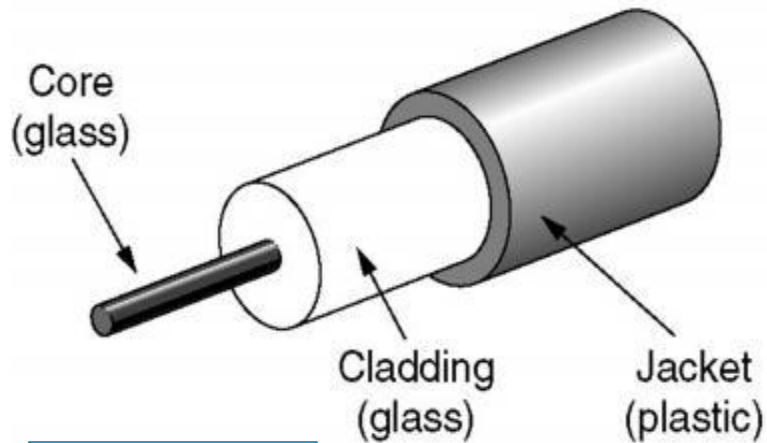
→ Terminators



- Terminators are required for bus topologies where one main cable acts as a backbone with branches to several devices but does not itself terminate in a device.
- If the main cable is left non terminated any signal transmitted over the line echoes. A terminator absorbs the wave at the end eliminates echo-back.
- It is used to prevent the signal disturbance.

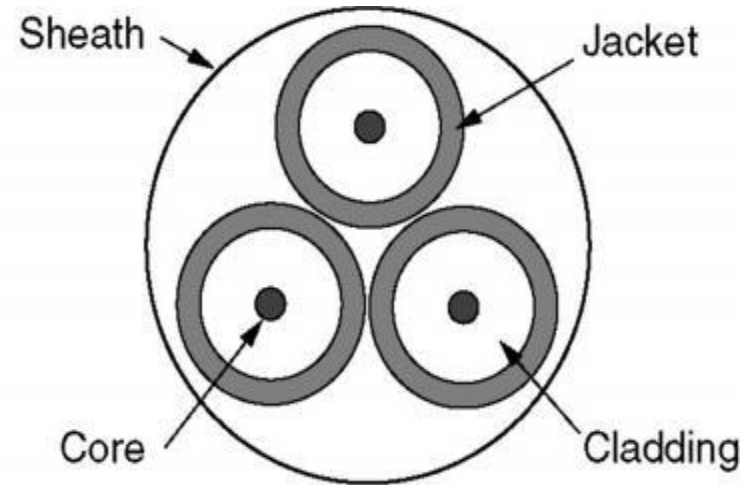


❑ Fiber Optic Cable (FOC)



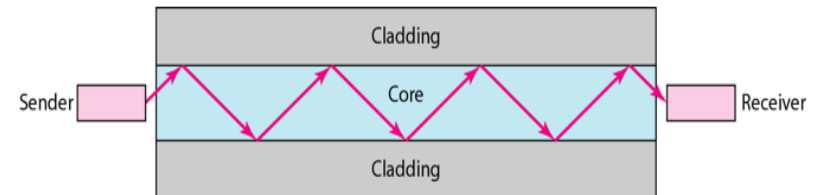
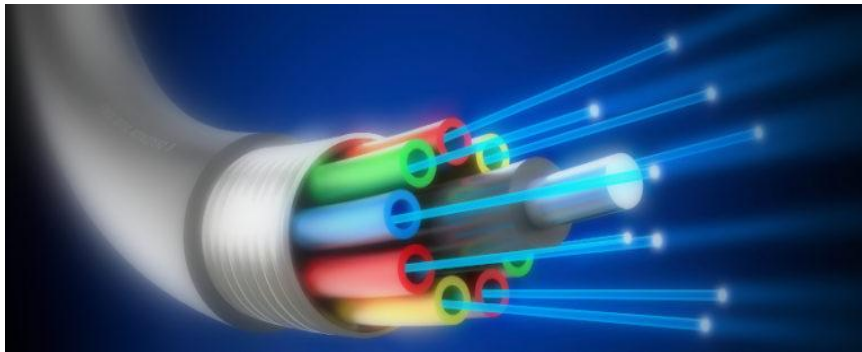
Side view of
FOC

(a)



End view of
FOC

(b)



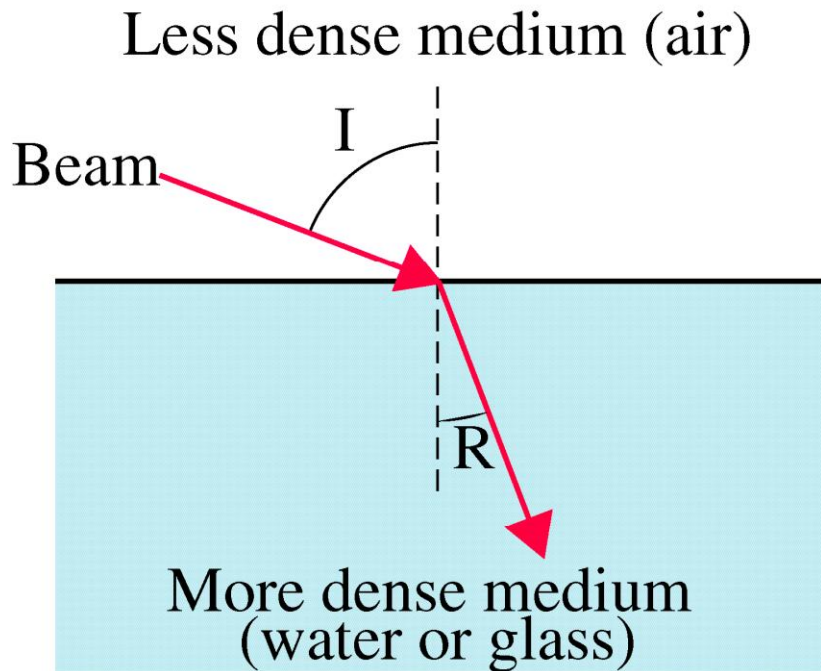
- Fiber optic cables are similar to coax, except without the braid.
- A fiber-optic cable is **made of glass or plastic** and transmits signals in the form of **light**.

- Optical fibers use reflection to guide light to move through a channel.
- As we can see in figure, a glass or plastic core is surrounded by a cladding of less dense glass or plastic.
- Figure shows a single fiber viewed from the side. At the centre is the glass core through which the light propagates.
- The core is surrounded by a glass cladding with a lower index of refraction than the core, to keep all the light in the core.
- Next comes a thin plastic jacket to protect the cladding. Fibers are typically grouped in bundles, protected by an outer sheath (પુસ્તક ઝાલકરણ). (that we can see in FOC figure b)
- Generally FOC comes in two types: Single mode and Multi mode. In single mode fibers → the core is about 8 to 10 microns and in multi mode fibers → the core is about 50 microns.

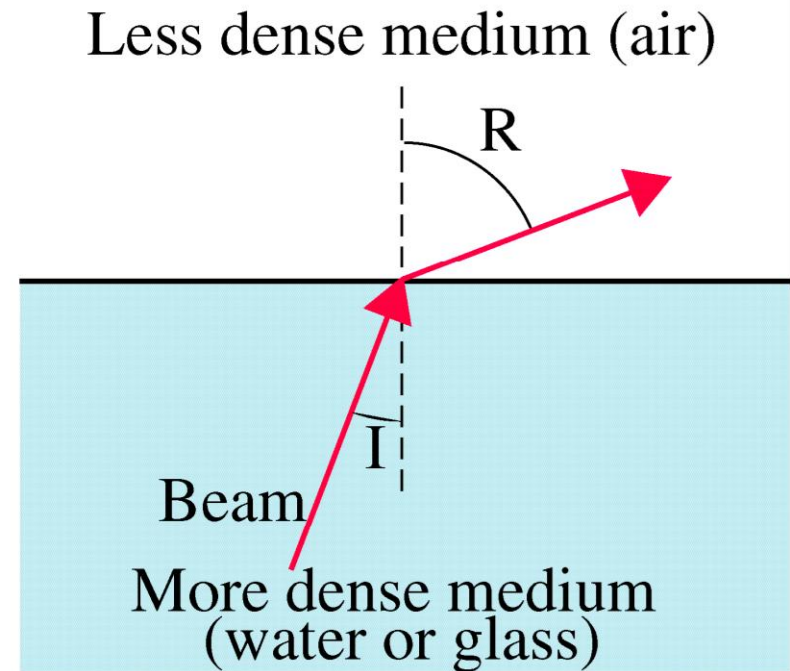
Concept of REFRACTION



Refraction

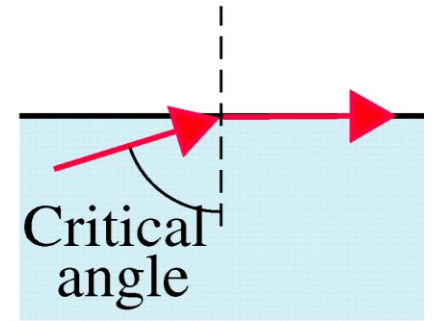
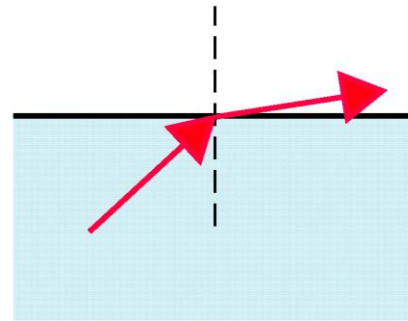
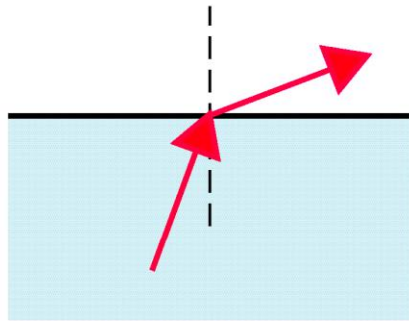
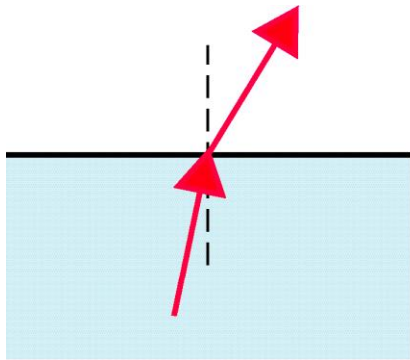


a. From less dense to more dense medium

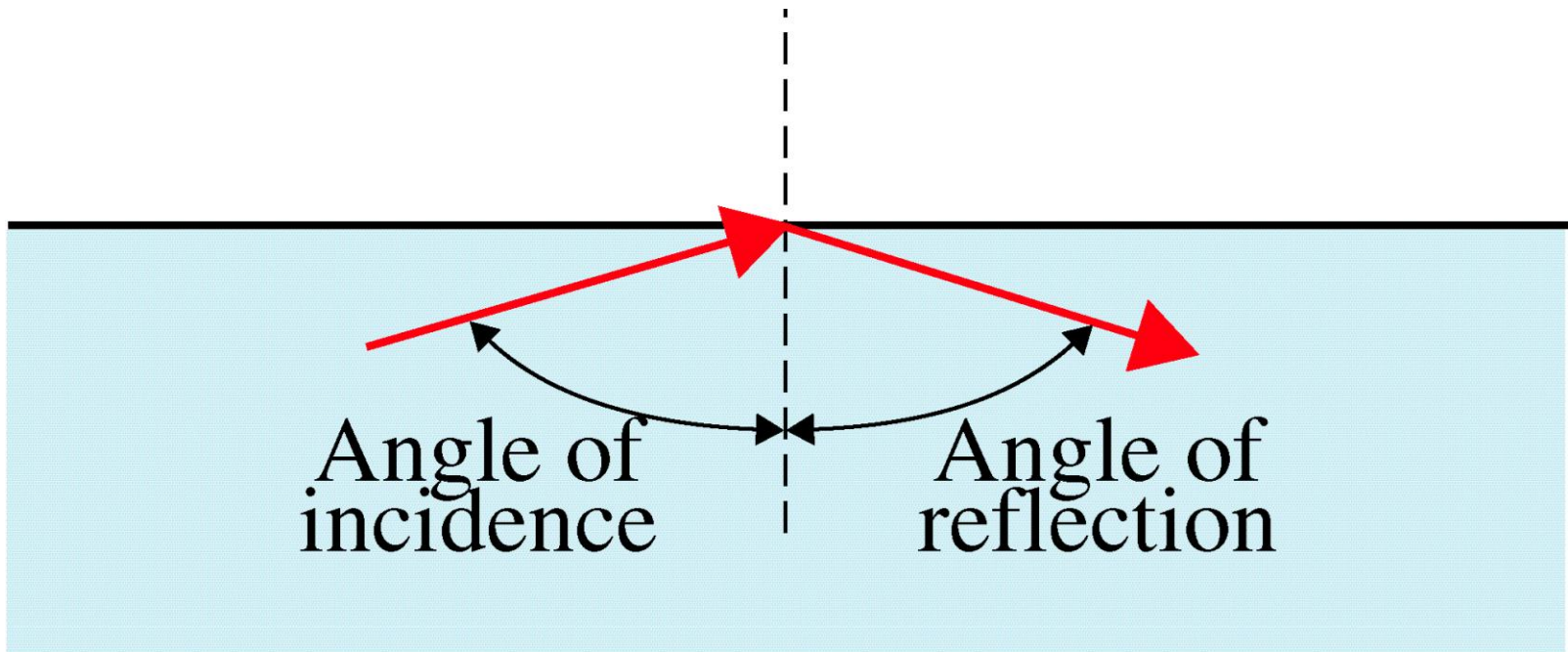


b. From more dense to less dense medium

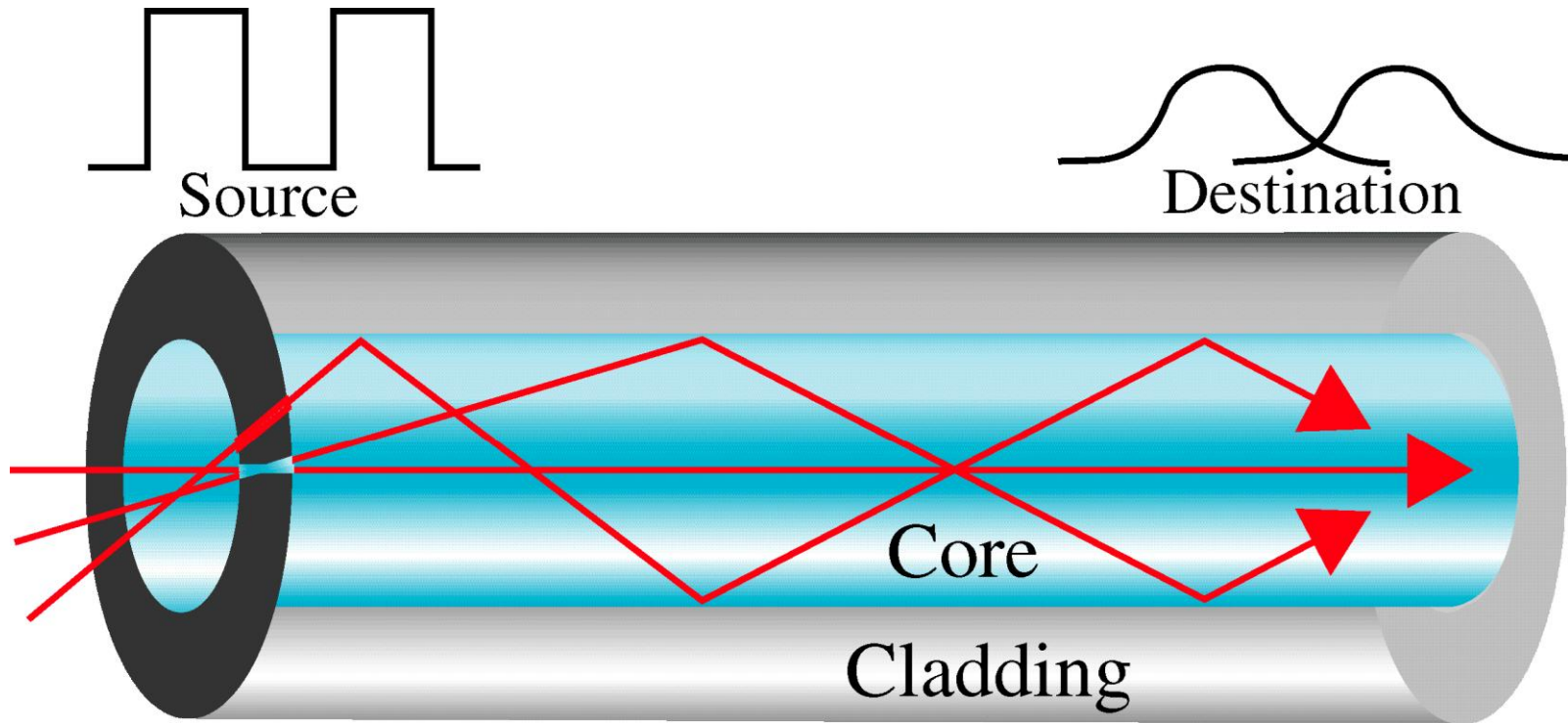
Critical Angle



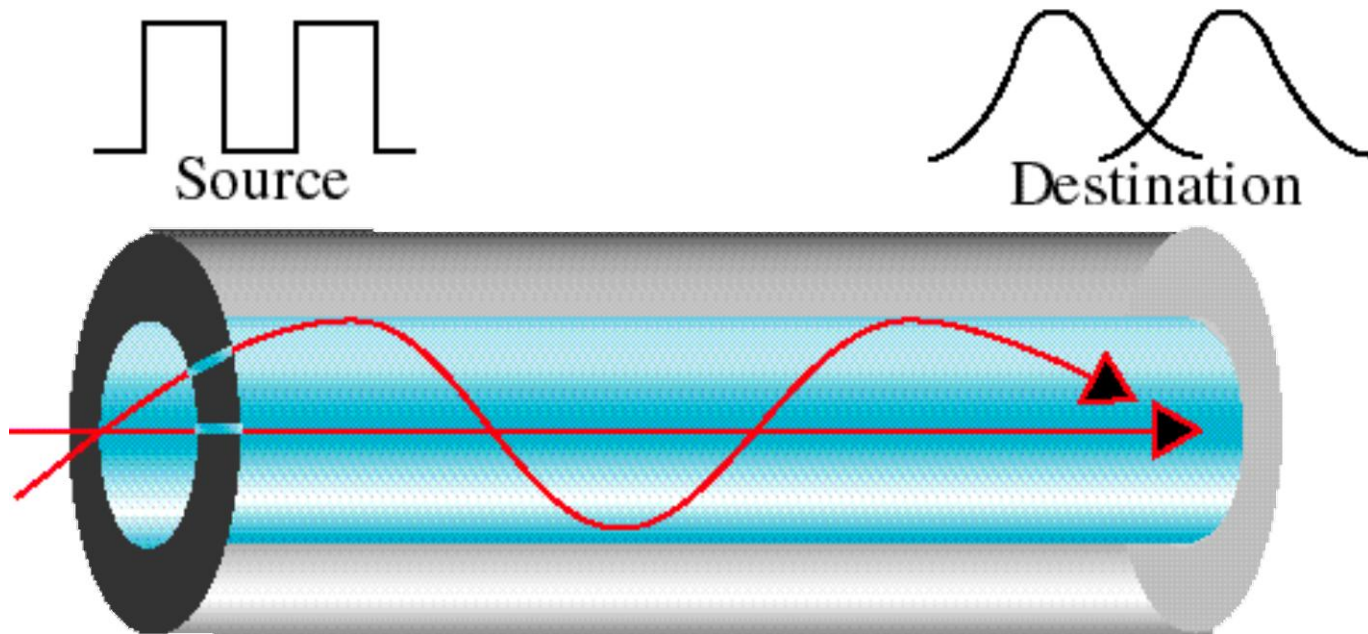
Reflection



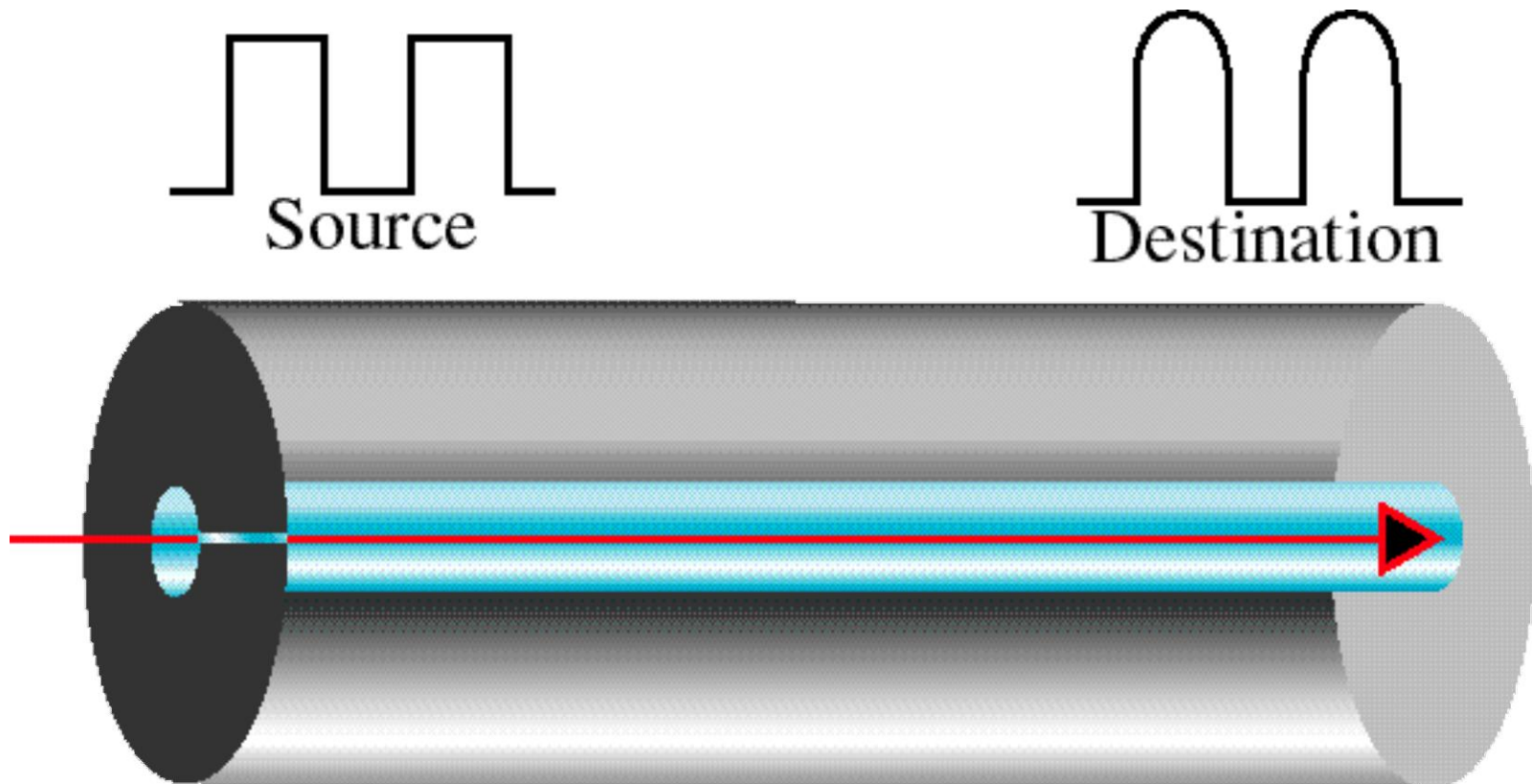
Multimode Step-Index



Multimode Graded-Index



Single Mode



➡ Advantages of FOC

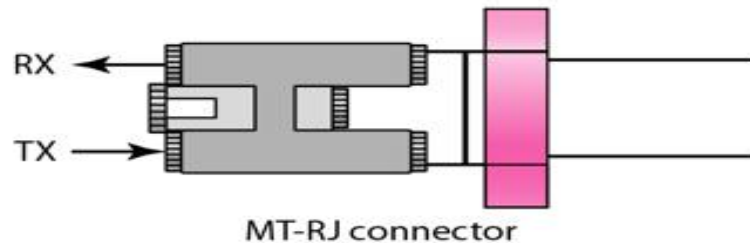
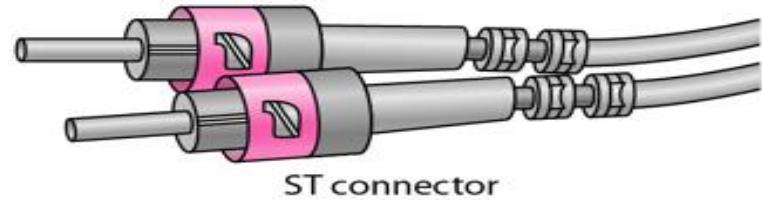
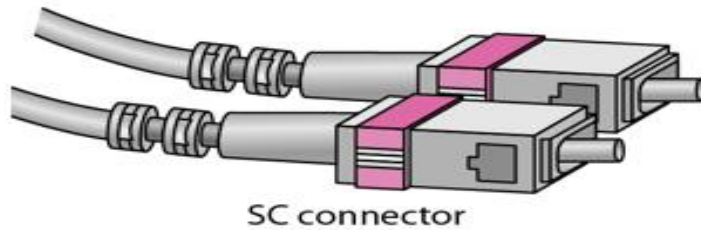
- *Higher bandwidth:* It can support dramatically higher bandwidths.
- *Light weight:* Fiber-optic cables are much lighter than copper cables.
- *Longer Distance:* in it, optical cables are capable of providing low power loss, which enables signals can be transmitted to a longer distance than copper cables.
- *Faster speed:* speed of data transmission is much higher in it.
- *Excellent Security:* it provides excellent security because they do not leak light and it is quite difficult to tap (વચ્ચેથી સંદેશો સાંભળવો).
- *Noise Resistance:* because fiber optic transmission uses light rather than electricity, noise is not damaging the signal.

- *Better reliability:* FOC is immune (ཡུན་རིང་ལྷོད་མེད་) to temperature changes, severe weather and moisture and it's not bothered about electromagnetic interference (EMI) which can interrupt the data.

➡ **Disadvantages of FOC**

- *Difficult to install:* special skills are required to install and maintain fiber optic cables.
- *Higher cost:* it is expensive compare to other cables.
- *Unidirectional (one direction):* propagation of light is unidirectional. If we need bidirectional communication then two fibers are needed.
- *Fragility:* it is made up with glass, so is can be easily broken and it is less useful for application where portability required.

♦ Fiber optic connectors



- Fiber optic cables use three different connectors as shown in above figure.
- 1. **The subscriber channel (SC):** connector is used for cable TV. It uses a push/pull locking system.
- 2. **The straight-tip (ST):** connector is used for connecting cable to networking devices.
- 3. **MT-RJ (Mechanical Transfer Registered jack):** is a new connector with same size as RJ45

→ Fiber optic communication

- Fiber-optic communication is a method of transmitting information from one place to another by sending pulses of light through an optical fiber.
- There are three main basic elements of fiber optic communication system.

1. Light source

Two components are used as light sources: light emitting diodes (LED's) and LASER diodes, in which LEDs are used for short distances and low data rate applications and LASER is used for longer distances and high data rate transmission.

2. Optical fiber as a medium

For data transmission, ultra thin optical fiber cable is used.

✓ Photo detector

The purpose of photo detectors is to convert the light signal back to an electrical signal. Two types of photo detectors are mainly used for optical receiver in optical communication system: **PN photo diode and avalanche photo diode.**

(Just for information)



Copper Twisted pair



Copper Coaxial Cable



Optic Fiber Cable

❖ Most common types of Ethernet cabling

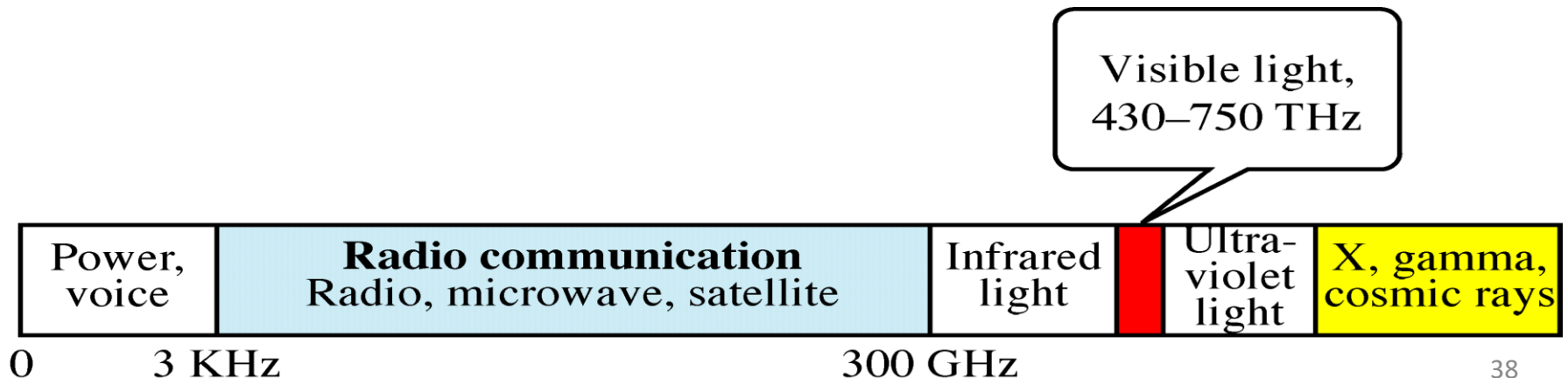
Name	Cable	Max Segment	Nodes	Advantages
10Base2	Thin Coax	185 m	30	No hub needed
10Base5	Thick Coax	500 m	100	Original cable, now obsolete
10BaseT	Twisted pair	100 m	1024	Cheapest system
10BaseF	Fiber optics	2000 m	1024	Best between buildings

LEARNING OUTCOME

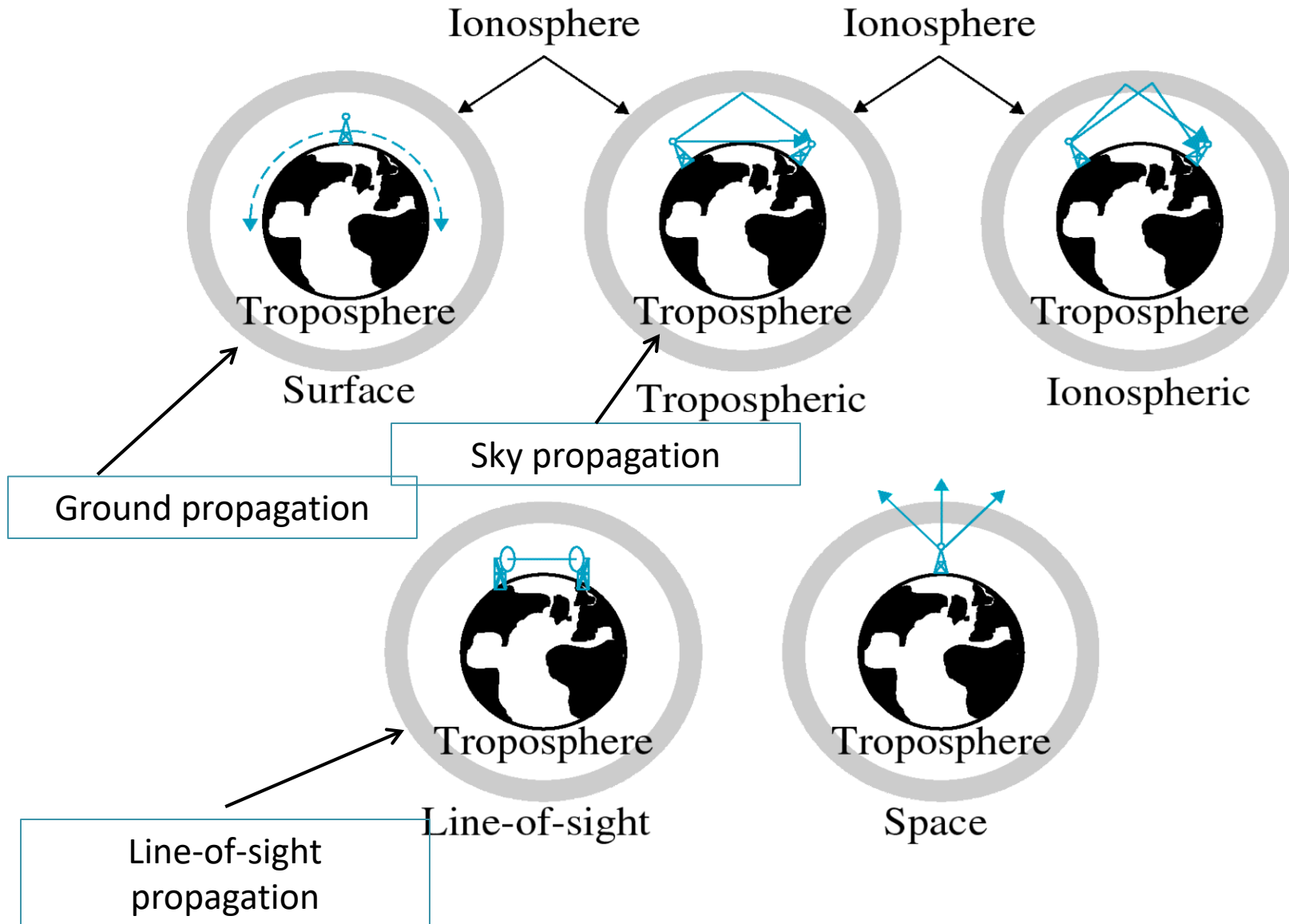
- After the completion of this session, students will be able to.... *select appropriate unguided transmission media for given network.*

3.3 Un guided media

- Unguided media transport electromagnetic waves without using a physical conductor.
- In it, signals are normally broadcast through air and thus are available to anyone who has a device capable of receiving them.
- This type of communication is often referred to as wireless communication.
- In some situations where it is impossible to use cables, unguided media are good alternatives.
- Electromagnetic spectrum for unguided media is shown below



→ Signal transmission in unguided media



→ Signal transmission in unguided media

- Unguided signals can travel from source to destination in several ways: *ground propagation*, *sky propagation*, and *line-of sight propagation*.

◆ Ground propagation

- In this propagation, radio waves travel through lowest portion of atmosphere, hugging the earth.
- These **low-frequency** signals **emanate (મિથી પસાર થવું)** in all **directions** from the transmitting antenna and follow the curvature of planet.
- Distance depends on the amount of power in the signal: **greater the power, the greater the distance.**

◆ Sky Propagation

- In this propagation, **higher-frequency** radio waves radiate upward into the ionosphere (the layer of atmosphere where particles exist as ions) where they are reflected back to earth.

- This type of transmission allows for greater distances with lower power output.

◆ Line of sight Propagation

- In this propagation, **very high-frequency** signals are transmitted in straight lines **directly from antenna to antenna**.
- Antennas must be directional, facing each other and either tall enough or close enough together not to be affected by curvature of earth.
- Line-of-sight propagation is tricky because radio transmission cannot be completely focused.

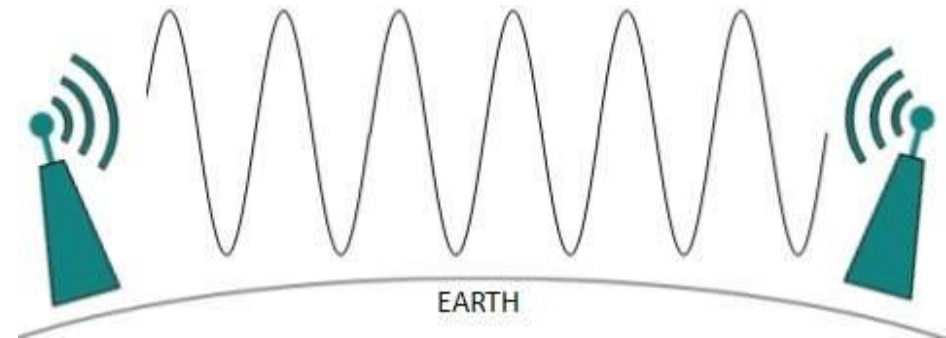
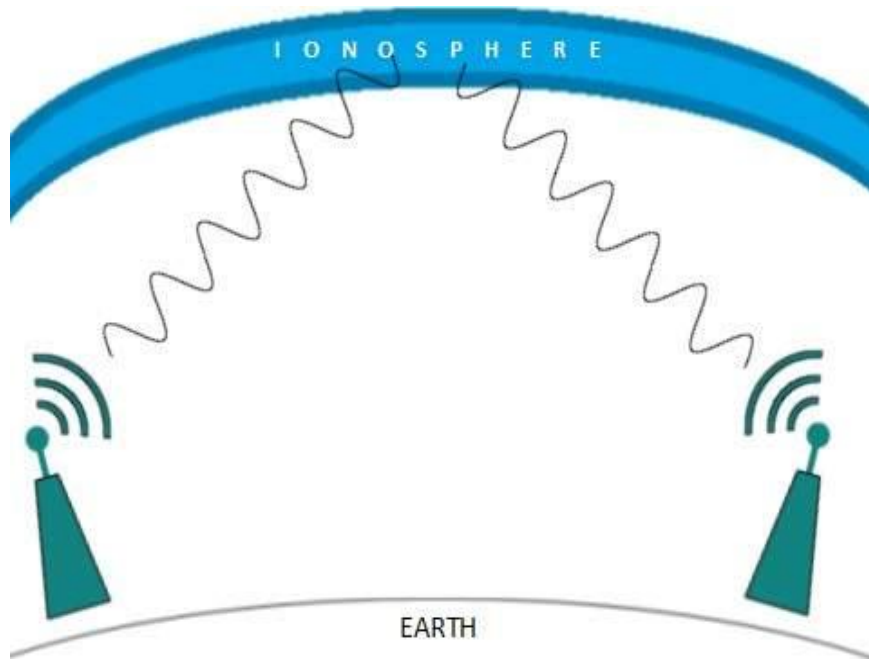
Frequency band for unguided media

Band	Range	Propagation	Application
VLF (very low frequency)	3 - 30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30 - 300 kHz	Ground	Radio beacons and navigational locator
MF (middle frequency)	300 kHz - 3 MHz	Sky	AM radio
HF (high frequency)	3 - 30 MHz	Sky	Citizens band (CB), Ship / aircraft communication
VHF (very high frequency)	30 - 300 MHz	Sky and Line-of-sight	VHF TV, FM radio
UHF (ultra high frequency)	300 MHz - 3 GHz	Line-of-sight	UHFTV, cellular phones, satellite
SHF (super high frequency)	3 - 30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30 - 300 GHz	Line-of-sight	Radar, satellite

♦ Classification of unguided media

- Main three categories are there → 1. Radio wave and Micro wave, 2. Infrared waves and 3. LASER

❖ Radio wave transmission



- Radio waves are electromagnetic waves occurring on radio frequency portion of the electromagnetic spectrum.

- These electromagnetic waves are ranging in frequency between **3KHz and 1GHz**.
- A common use is to transport information through the atmosphere or outer space without wires.
- Radio waves are **Omni directional**, means they **can travel in all direction from the source**. So transmitter and receiver **do not have to be aligned physically**.
- Radio waves are easy to generate, can travel long distance and can penetrate (મિથી પસાર થવું) buildings. So they are widely used for communication **in both indoors and outdoors**.
- Radio waves are of 'sky propagation'.

↪ **Advantages of radio waves**

- It can travel long distance (for ex. AM radio).
- Widely used for communication in both indoors and outdoors.
- Radio waves can travel in omni direction, so no need to aligned antennas physically.

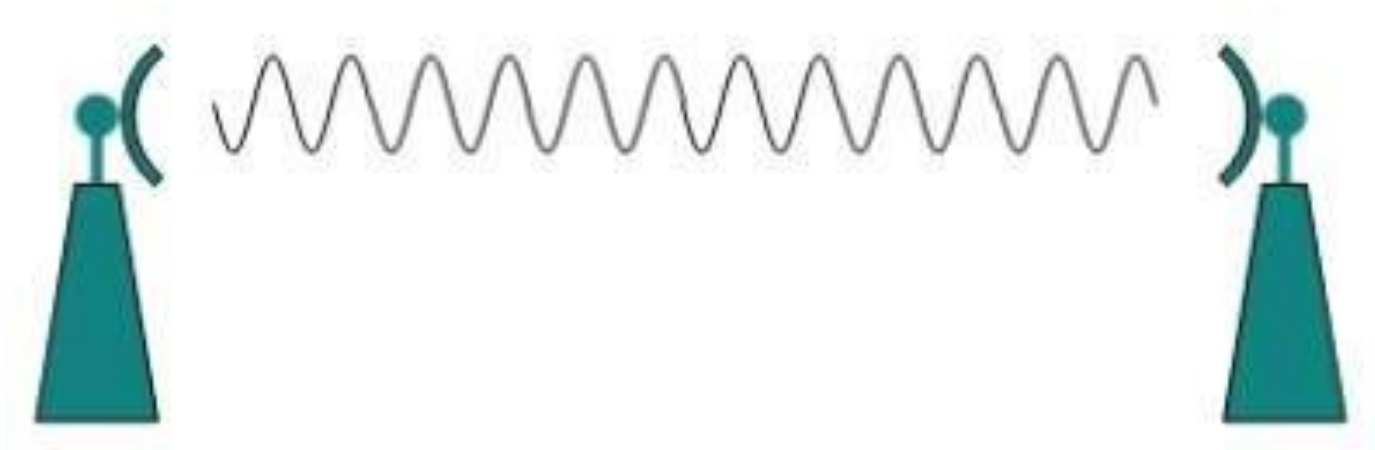
↪ **Disadvantages of radio waves**

- Because of omni directional antenna, the signal sent by one antenna can be interfered by another antenna that may be using same frequency or band.
- They can penetrate walls, so we can not isolate whether the communication to just inside or outside the building.
- Due to the narrow side bands (up to 1GHz), there is low data rate for digital communication.

↪ **Applications of radio waves**

- AM and FM radio
- GPS service
- Cordless phone
- Police radio
- Television broadcast.

❖ Micro waves transmission



- These electromagnetic waves are ranging in frequency between **1GHz and 300GHz**.
- These waves are **unidirectional**, means when an antenna transmits microwaves, the sending and receiving antenna **need to be aligned**. Signals of micro waves are **narrowly focused**.
- Micro waves are of 'line-of-sight' propagation.

↪ **Advantages of micro waves**

- There is no interference of another pair of aligned antennas.
- As **band is wider than radio waves**, high data rate for digital communications.
- Inexpensive and easy to install.

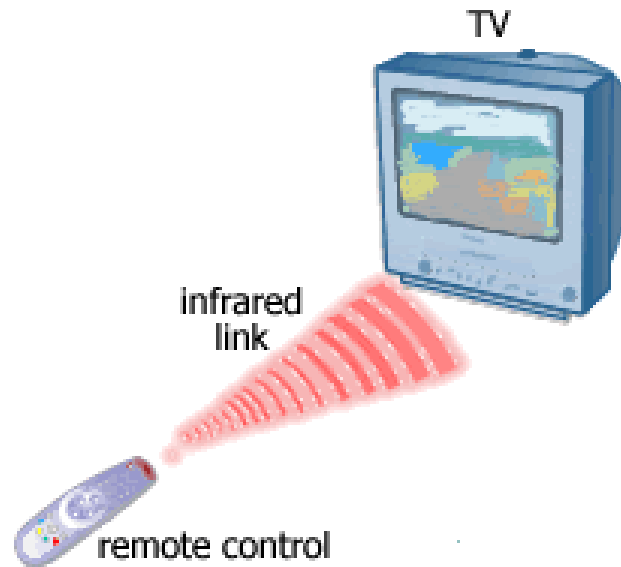
↪ **Disadvantages of micro waves**

- These waves are unidirectional, so sending and receiving antennas must be aligned.
- Repeaters are required for longer distance.
- Very high frequency waves can not penetrate the walls, so receivers inside the buildings can not communicate.

↪ **Applications of micro waves**

- These waves are used for unicasting (one sender and one receiver).
- Cellular phones
- Satellite networks
- Wireless LANs

❖ Infrared waves transmission



- These electromagnetic waves are ranging in **frequency between 300GHz and 400THz**.
- These waves are **used for short range communication**.
- Examples of infrared are → remote controls used on televisions, VCRs, and stereos.
- Infrared waves are of 'line-of-sight propagation'.

↳ Advantages of infrared waves

- Infrared waves are **relatively directional**, cheap, and easy to build.
- Infrared waves having **high frequency**, so they are **not able to penetrate the walls**, so in a room infrared systems can't be interfered.
- Infrared waves are useful for short range communication. It is more secure against taping.

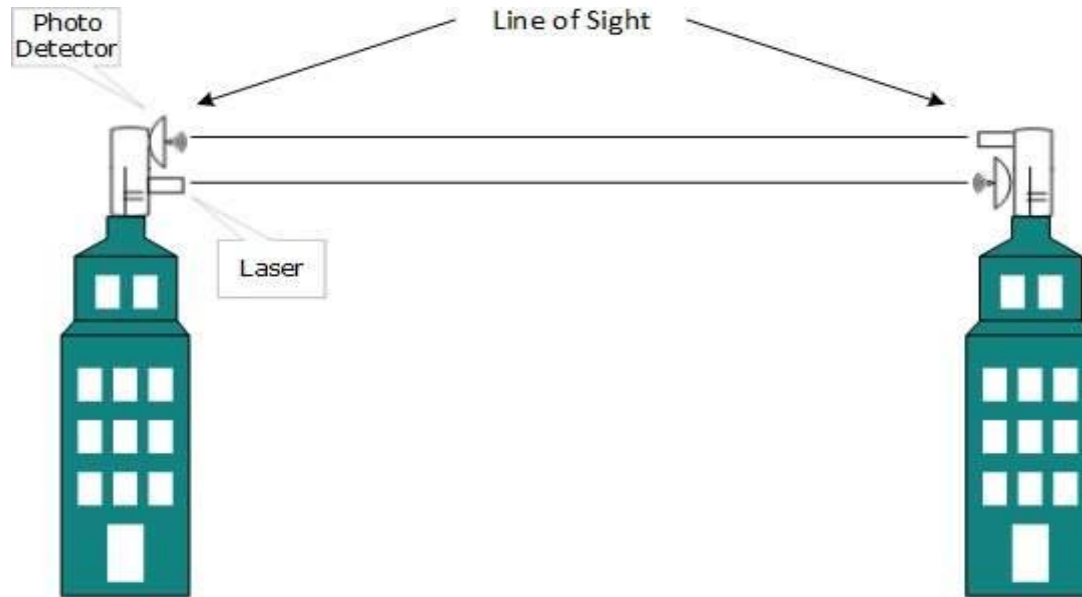
↳ Disadvantages of infrared waves

- These waves are useless for long range communication.
- They **do not pass through solid objects**.
- Sun's ray can interfere the radio waves, so it is not good to use infrared outside the buildings.

↳ **Applications of infrared waves**

- Used in short range wireless communication devices (e.g. Wireless mouse or keyboard).
- Used in remote controls of TV, VCRs and stereos based on infrared.
- Used in thermal efficiency analysis, remote temperature sensing, weather forecasting.
- Also used in military applications: target acquisition, surveillance, tracking.

❖ LASER transmission



- The full form of LASER is → Light Amplification Stimulated Emission of Radiation.
- Laser beams are unidirectional, therefore this type of transmission system use 'line-of-sight' propagation. So sender and receiver must be aligned
- It is similar to infrared technology but can transmit signal to long distance.

- *Lasers cannot penetrate obstacles such as walls, rain, and thick fog. Additionally, laser beam is distorted by wind, atmosphere temperature, or variation in temperature in the path.*
- In such a transmission system, *a photo detector and laser is set on both sender and receiver side.*
- The carrier used for the transmission signal is typically generated by laser diode.

➔ **Advantages of laser**

- Laser is safe for data transmission as it is **very difficult to tap 1mm wide laser** without interrupting the communication channel.
- It offers very high band width at a very low cost.
- It is free from electro-magnetic interference. This phenomenon is used in optical wireless communication through free space for telecommunication as well as computer networking
- High speed.
- It has very minimum signal leakage.

- Higher data storage is possible because single laser beam can be focused in areas smaller than 1 micro diameter.

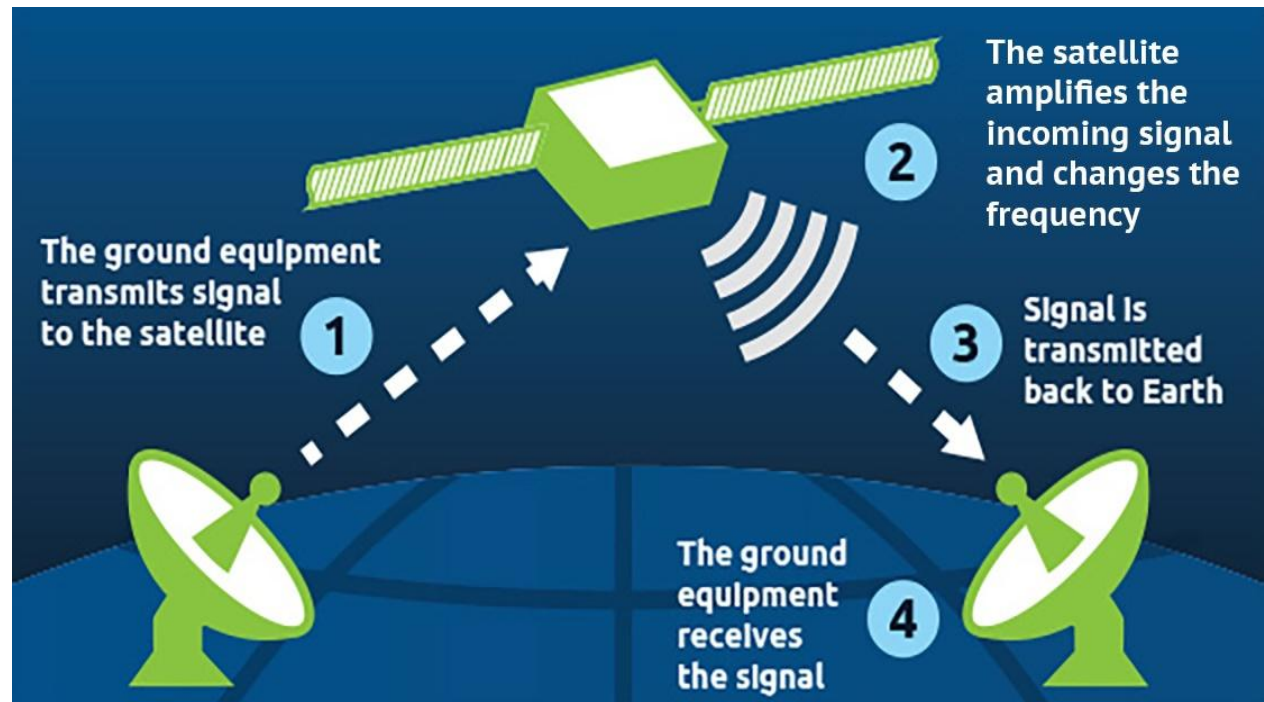
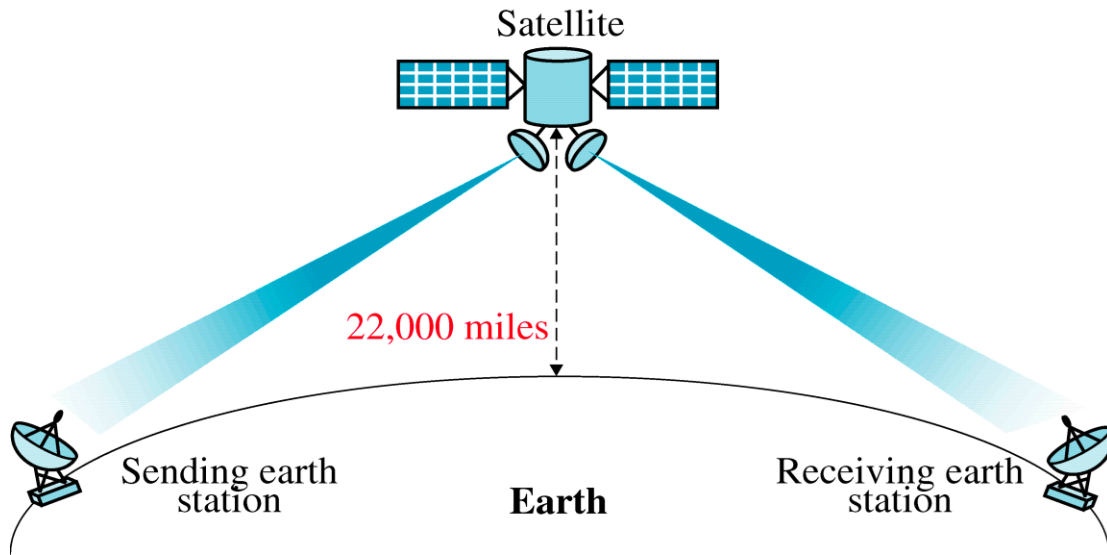
↳ **Disadvantages of laser**

- Maintenance is costly and difficult.
- It is harmful to human beings and often burns them during contacts.

↳ **Applications of laser**

- Laser communications have been utilized for mass communications including telephone conversations and even television channels.
- Used in digital data transmission.
- Effectively used for shorter distance communication.

❖ Satellite communication



- The satellite is man made equipment that is placed in orbit around the earth. They provide communication from one point on the Earth to another.
- Satellite work as transmitter/receiver that is launched by rocket and placed in orbit.
- Satellites enable communication over large distance (beyond line of sight).
- Used in communication in ships, aircrafts and other mobile vehicles, weather forecasting, television broadcast, internet communication, radio communications and global positioning system (GPS).
- There are two types of communication satellites → ACTIVE and PASSIVE.
- Active satellite acts as a repeater, it amplifies the received signals and retransmits them back to earth.

- Passive satellite only reflects received radio signals back to earth.
- As we have seen in the figure, earth station transmits signals to the satellite using UP-LINK frequency and the satellite then amplifies the signals, convert them and transmit back to the receiving earth station using DOWN-LINK frequency.

↳ **Advantages of satellite communication**

- Mobile communication can be easily established by satellite communication.
- It is best communication for long geographical distance where cable communication is impossible.
- Effective point-to-multipoint communication.
- Signal quality is high compared to fiber optics.
- Low operating cost.

↳ **Disadvantages of satellite communication**

- Delay in signal transmission.
- Efficiency reduced in long file transfer.
- Manufacturing of satellites is expensive.

↳ **Applications of satellite communication**

- TV/radio broadcasting and global mobile communication.
- Weather forecast, military, navigation

Thank YOU...

Unit : 4

Network Devices

Prepared By:

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Lecturer, AVPTI, Rajkot

COURSE OUTCOME (CO4)

- After completion of this Unit, students will be able to ***Illustrate use of various network devices used in computer network.***

LEARNING OUTCOME (LO)

- After completion of this session, students will be able to *explain use of various network devices*

↳ Introduction

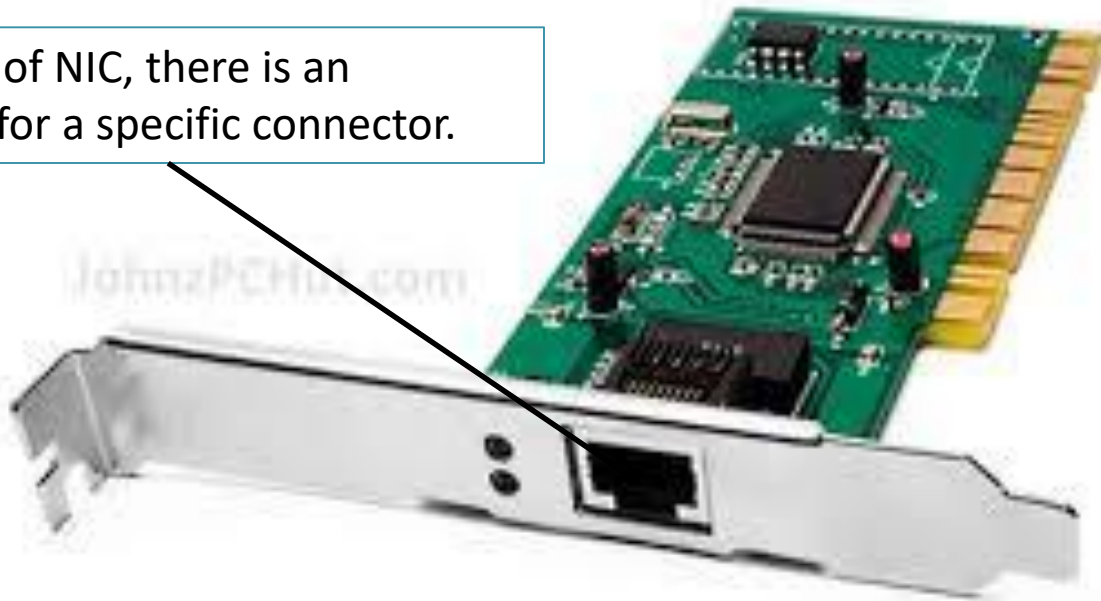
- Two or more devices connected with each other for sharing their data and resources are called a network.
- When two networks are connected, then called internetwork.
- Examples of networking and internetworking devices are:
 - Repeater, Hub, Switch, Bridge, router, gateway etc..

Network Adapter (NIC)

- It is ad-on card available on motherboard.
- It is a small printed circuit board that is installed on a mother board of CPU.
- This is additional hardware card required to install physically on the slot of motherboard.

- This card provides the connectivity between computer internal resources and external resources connected to the network.
- Each NIC has a unique address called MAC address / physical address that is different from other NIC.
- Types of NICs are
 1. **ARCNET Card**
 2. **Ethernet Card**

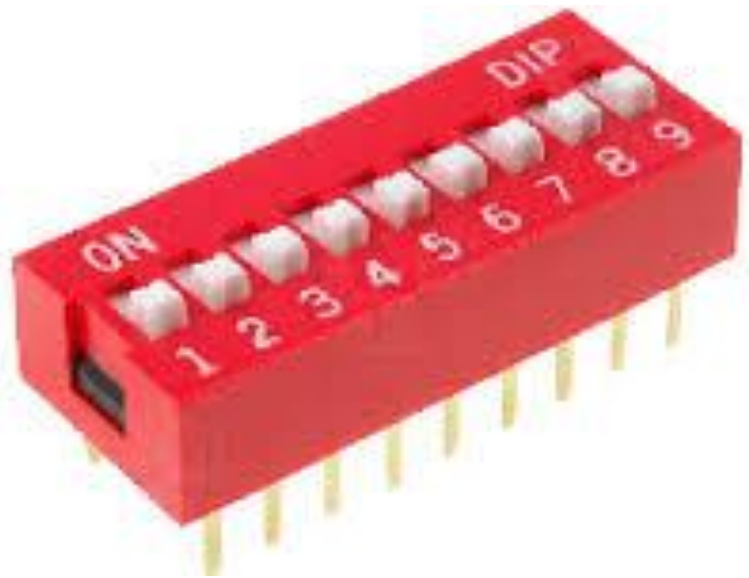
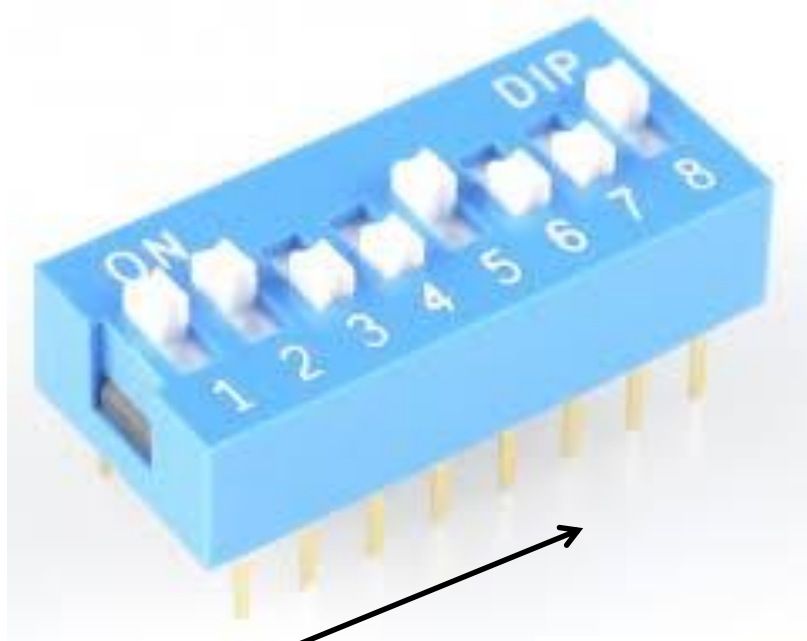
Back side of NIC, there is an interface for a specific connector.



NIC

1. ARCNET Card

- The full form of ARCNET is **A**ttached **R**esource **C**omputer **N**etwork.
- It consists of 8 dual inline packages (DIP) switches. Which are used for defining address of nodes.
- If all switches are OFF then the address is 00h and if all ON then address is FFh.
- So we can connect 256 nodes with it.
- It supports MCA (micro Channel Architecture) bus.
- The cable used for this card is co-axial (RG-62).
- It has 8/16 bit port.
- Its transmission rate is 2.5 mbps.
- Has BNC connector
- ARCNET uses token ring bus access method.
- This type of NIC supports BUS topology and STAR topology.



DIP switches



ARCNET

2. ETHERNET Card

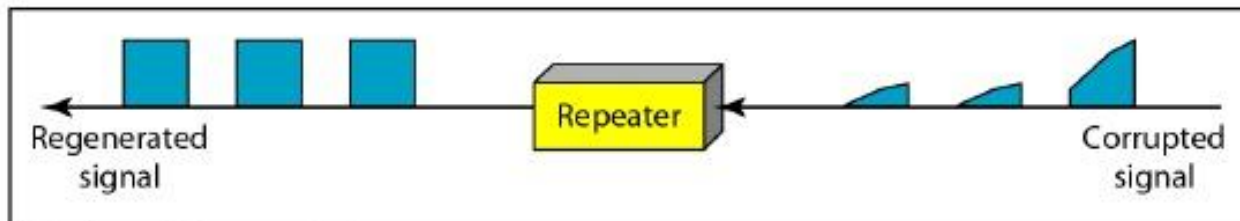
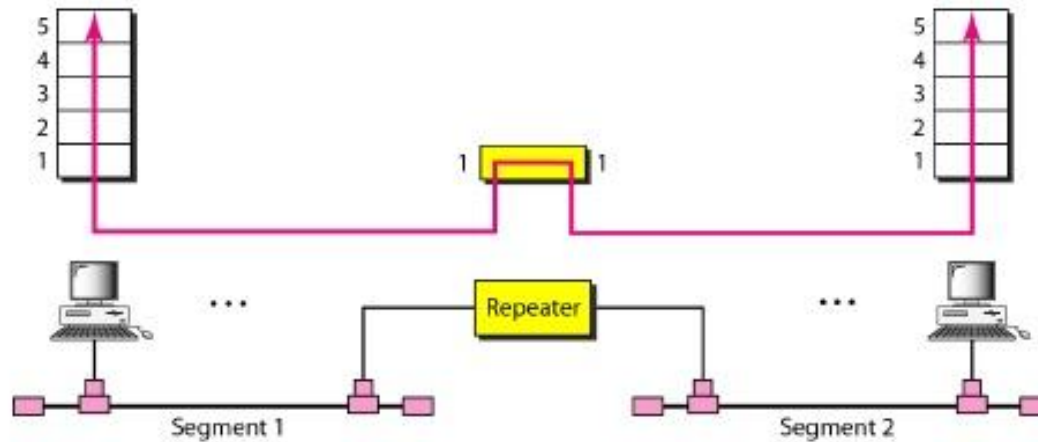
- It consists of PCI (peripheral component interface) bus.
- The slots available for this card are 8,16,32 bits.
- Node address is between 1 to 1023 (total 1024).
- DIP switches are not available.
- Cable used: RG-58 (previously) UTP (Now a days)
- Data transfer rate is 10 mbps.
- It uses BNC(previously) and RJ-45 (Now a days) connector.
- No switches for port address settings available.



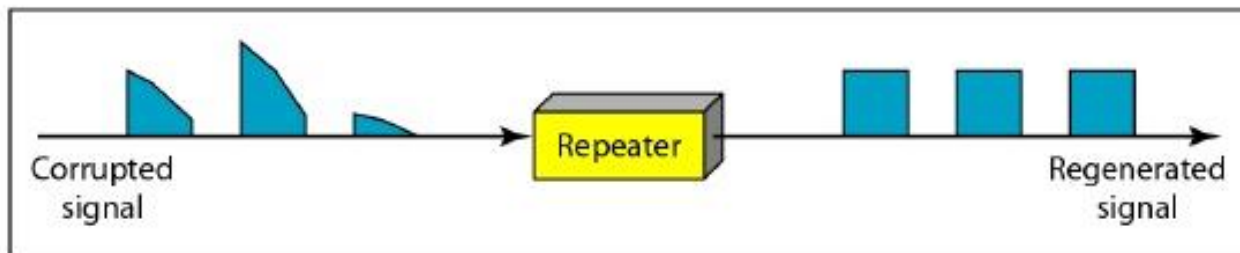
Repeaters

- Repeater is an electronic device which operates only in physical layer.
- Signal that carry information within a network can travel a fixed distance before noise can affect its integrity.
- Repeaters, installed on a link receives the signals before it becomes too weak or corrupted, it regenerates the original bit pattern and puts the new refreshed copy of signal back on the link.
- Repeaters allows us to extend only the physical length of a network.
- it does not change the functionality of the network.
- We can see the functions of the repeater in below figure.

Repeater connecting two segments of a LAN



a. Right-to-left transmission.

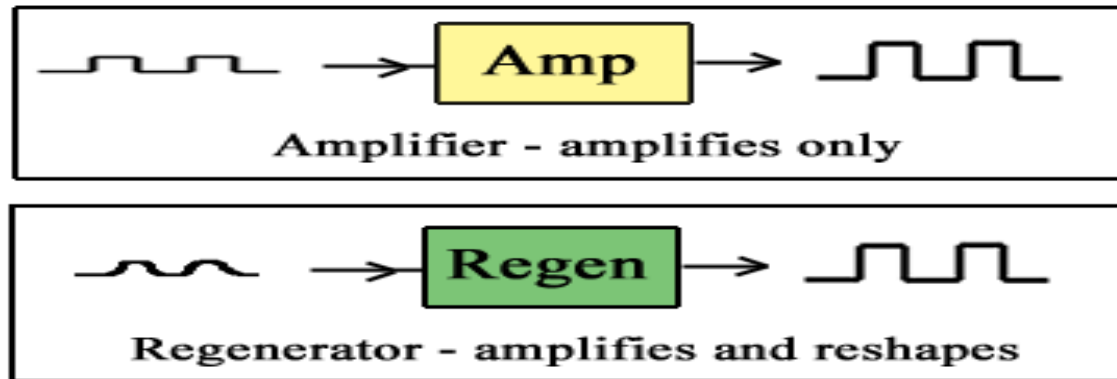


b. Left-to-right transmission.

Function of a repeater

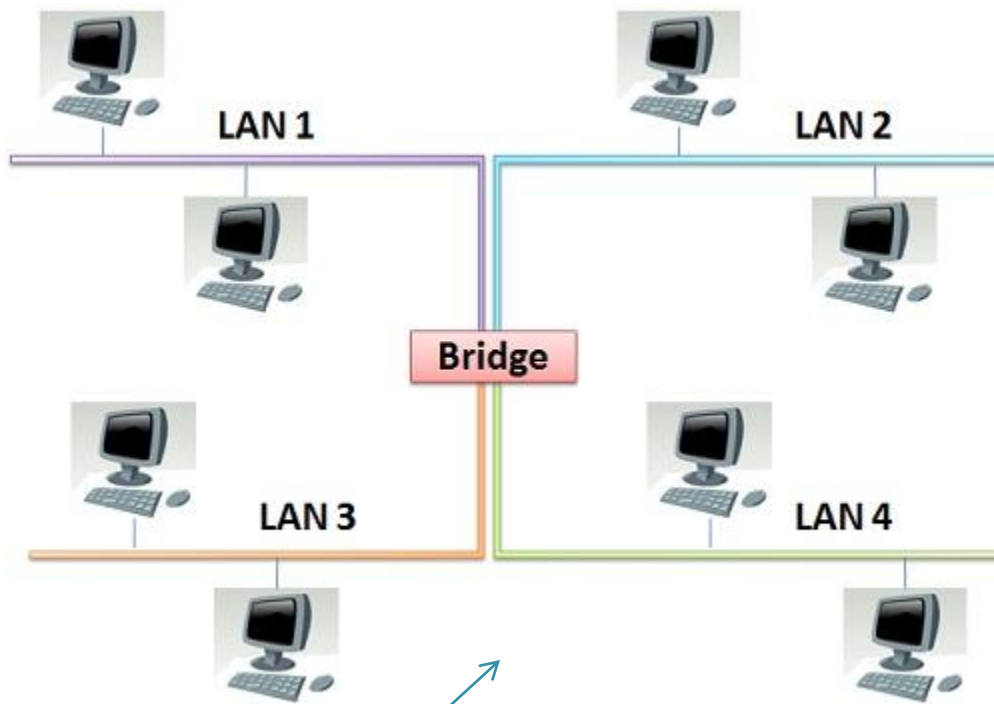
➔ Repeater is not an amplifier-JUSTIFY

- An amplifier can not differentiate the intended signal and noise signal, hence it amplifies equally everything fed (given as input) into it.
- *While repeater does not amplify the signal instead it regenerates it.*
- Repeaters placed so that a signal reaches it before the noise changes the meaning of any of its bits. So repeater is placed on the link before it becomes the weak or corrupted.



Bridge

- It operates on both physical and data link layer of OSI model.
- It connects two or more LANs.
- Bridge can divide the large network in to smaller segments.
- *Bridge has a software which keeps the separate traffic for each segment.*
- *Repeater transmits frames to entire connected devices while bridge transmits the frames only to separate segments.*
- Like this way, it **filters the traffic**.
- Bridges work on data link layer hence gives the access to physical address of the stations.
- Bridge contains the **look up table** that contains the physical addresses of every stations connected to it.
- Bridge is also worked for collision resolution.

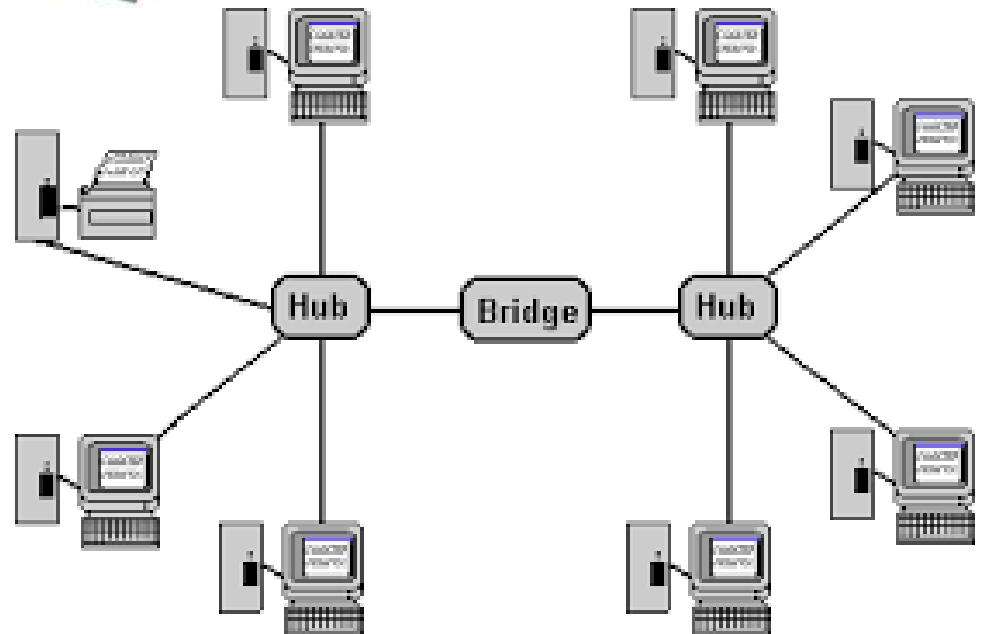


An Ethernet Bridge Connecting Two Segments

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Bridge placed in a network between different segments



- In the working of a bridge → when a frame enters to the bridge, it regenerates the signal and it checks the address of the destination and forwards the new copy only to the segment to which the address belongs.
- As a bridge found the frame, it reads the address contained in the frame and compares the address with a table of all the stations on both the segments.
- When bridge finds the correct match, it finds to which segment the station belongs and send the frame to only that segment.

↪ **Types of Bridges**

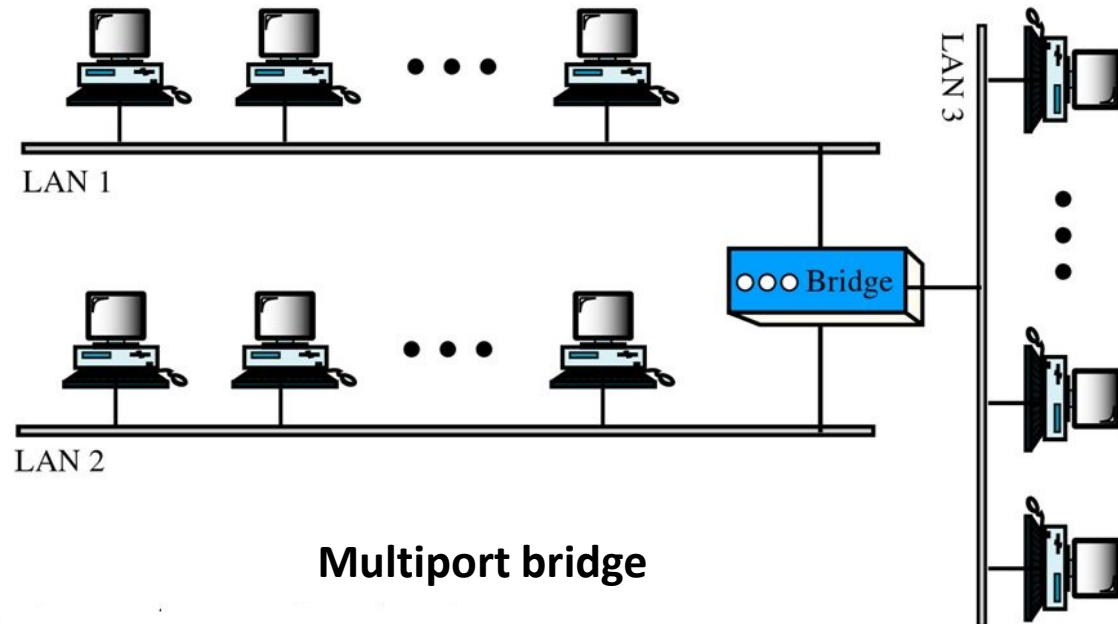
- 1. Simple bridge**
- 2. Multiport bridge**
- 3. Transparent bridge**

1. Simple bridge

- It is the most primitive (મૂળ, પ્રાચીન ઢબનું) and least expensive types of a bridge.
- It links two segments.
- It contains a table that lists the physical address of all the stations connected with it. Physical address must **entered manually**.
- In this bridge, **updating of device is time consuming** when new device is added / removed, the table must be modified at this time.
- So installation and maintenance for this bridge is time consuming and tedious.

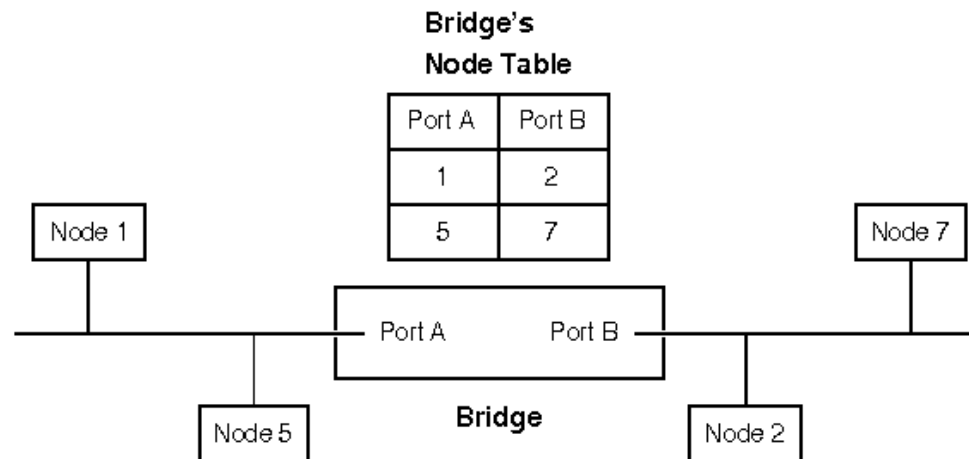
2. Multiport bridge

- It is used to connect more than two LANs.
- In this bridge, different tables are created each one holding the physical address of stations reachable through the corresponding port.



3. Transparent bridge

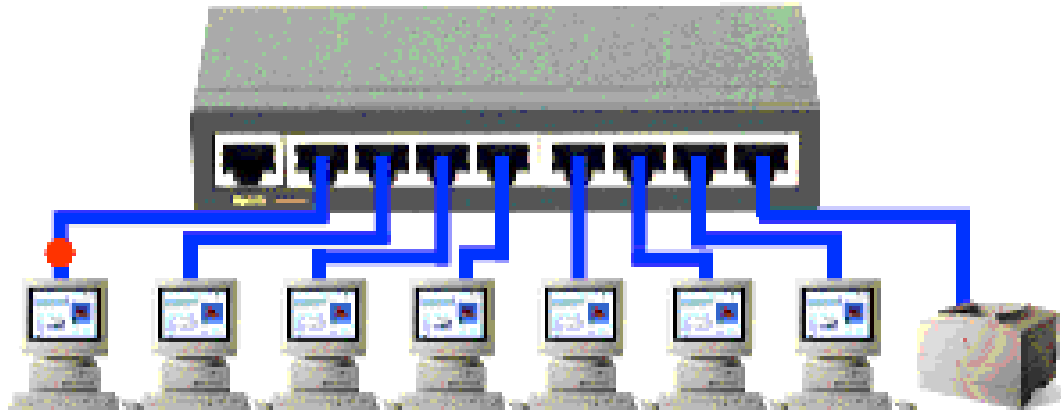
- It builds the table of physical address on its own as it performs its bridge function.
- Table is automatically built as frames are moved in the networks.
- **Initially** when bridge is installed, it is **empty**.
- As transmission is done the table is updated accordingly.
- The first packet transmitted by each station, the bridge makes entries inside the table with corresponding segment. So at last the table is completed with all details.
- So we can say, this bridge has **‘self updating features’**.



Hub

- Hub is used to create connections between stations in physical star topology.
- Large number of computers can be connected in single or multiple LANs.
- Hub is a central network device that connects network nodes. So is also referred as concentrator.
- Hub enables central network management.
- It provides connection for several different media types like: coaxial, fiber optic, twisted pair.
- Hubs are available in 8/16/24 ports.
- It provides high speed communication.

Hub



↪ Types of Hub

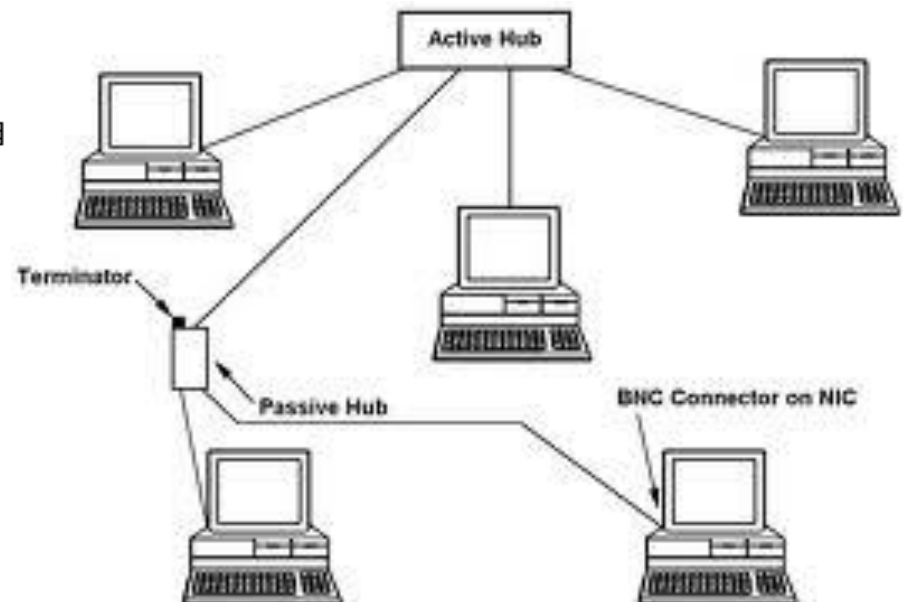
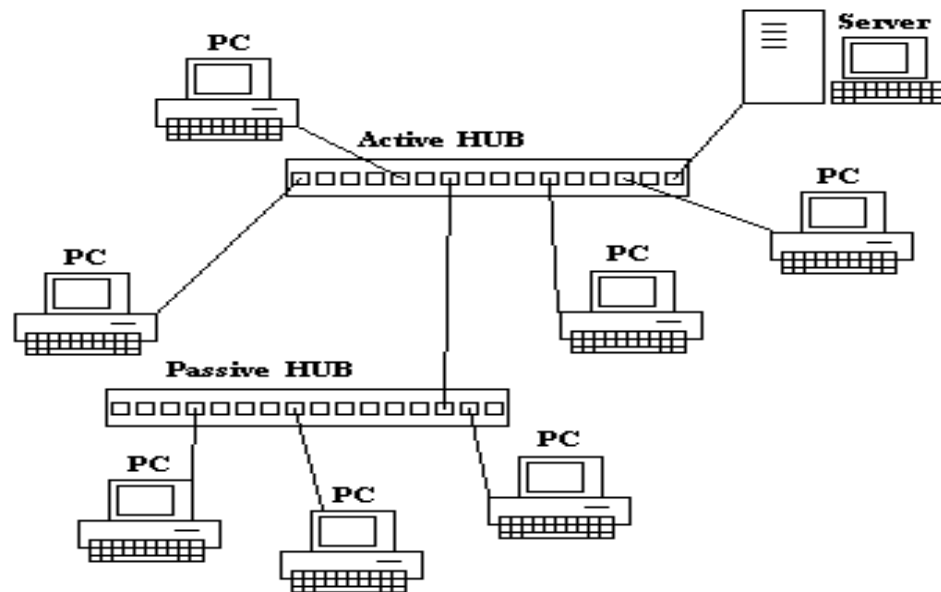
1. Active hub

2. Passive hub

3. Intelligent hub

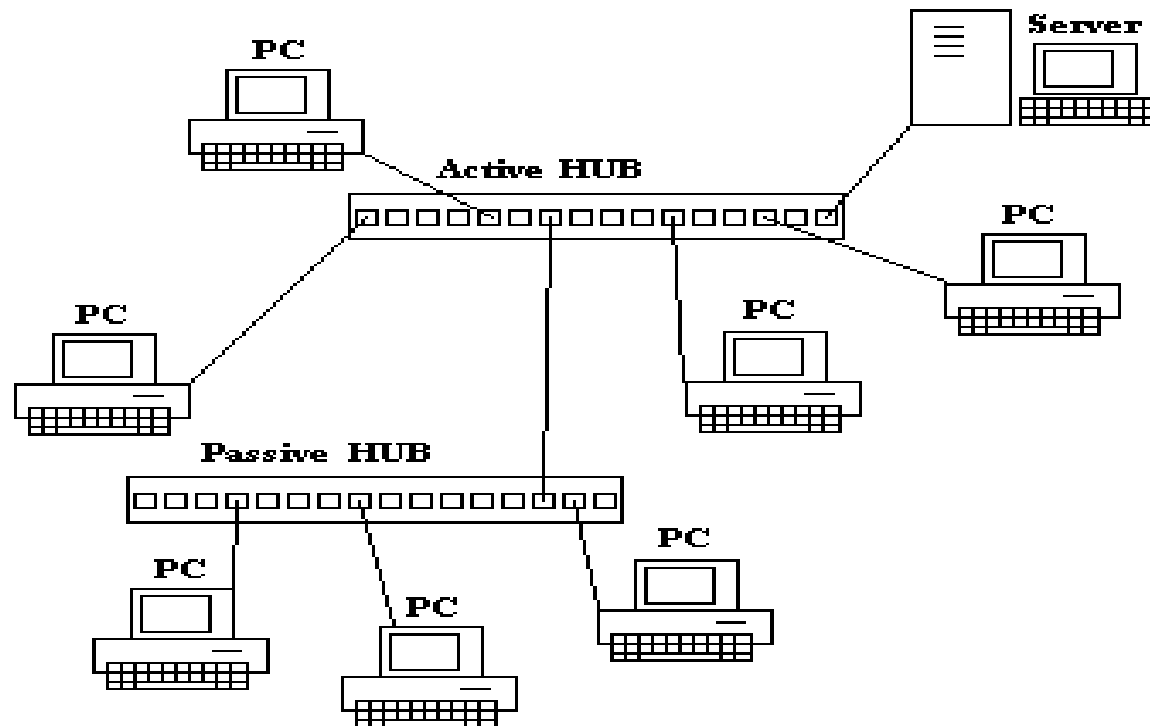
❖ Active hub

- It acts like repeaters, it regenerates and retransmits the signals.
- Because of multiple ports are available in it, it also called ***Multiport Repeaters.***
- it needs electric power to run.



❖ Passive hub

- It provides the physical connections between the attached devices.
- It do not amplify or regenerates the signals passes through hub.
- It do not require electricity to run.



❖ Intelligent hub

- It is the third and last type of hub which can perform task of both active and passive hub.
- It can accommodate several types of cables.
- It can perform some other tasks like Bridging and routing.
- It increases the speed and effectiveness of total network thus makes the performance of whole network fast and efficient.
- It is intelligent because capable of doing hub management and switching functionality.

➤ Applications of hub

- Hubs are used to create small Home Networks.
- Hubs are used for monitoring the networks.
- It Makes one device or peripheral available throughout the whole network.
- Hubs are used in Organizations and Computer Labs for connectivity.

Switch

- Switch provides the bridging functionality with greater efficiency.
- It acts as a multiport bridge to connect different devices or segments in a LAN.
- It operates in data link layer of OSI model (also called layer 2 switch).
- Switch has **buffer** for each link connected with it.
- **When it receives the frame, it stores the frame in the buffer of receiving link and checks address to find outgoing link.**
- If outgoing link is free the switch sends the frame to that particular link.
- two different characteristics of switch:
 - **Store and forward switch** → it stores the frame in input buffer until the whole frame is arrived.
 - **Cut through switch** → it forwards the frame to the output buffer as soon as the destination address is received.

LEARNING OUTCOME (LO)

- After completion of this session, students will be able to *Differentiate Layer 2 and Layer 3 Switches.*

Difference between layer 2 and layer 3 switch

Layer2 switch



Layer3 switch



❖ Layer2 switch

- It operates using **MAC address** for sending the frames from one node to next node.
- It performs at the **physical and data link layer** of OSI model.
- L2 switch is like a multiport transparent bridge, which allows better and faster performance.
- L2, as a bridge does the filtering decision based on the MAC address of the frame it received.
- It has **buffer to hold the frames** for processing.
- It has the switching factor that can forwards the frame faster.
- Generally it is used in **small private network**.
- L2 provides the following benefits:

- Hardware based bridging	- High speed
- Low latency	

❖ Layer3 switch

- It is used at the network layer of OSI model. (layer-3, hence called layer-3 switch).
- L3 switch is like a router, but it is more faster and sophisticated.
- Switching in L3 switch allows faster table lookup and forwarding.
- It operates utilizing IP address to make forwarding decision.
- L3 is best for large network and working good in case of heavy traffic.
- L3 switch provides following benefits:

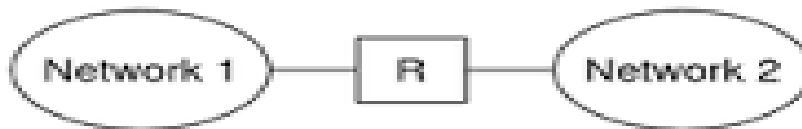
- High speed scalability	- High performance packet switching
- Low latency	-Security
- Quality-of-Service (QoS)	- Flow accounting

LEARNING OUTCOME (LO)

- After completion of this session, students will be able to *explain use of various internetworking devices*

Routers

Two Networks Connected by a Router



Three Networks Connected by Two Routers



Figure 1

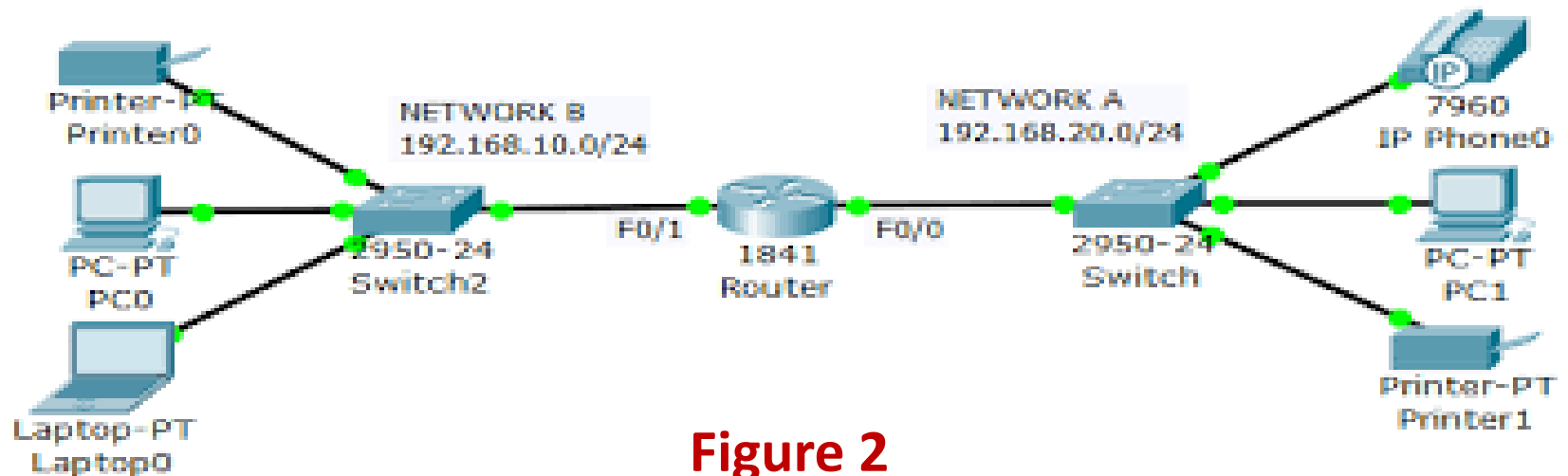


Figure 2

Routers

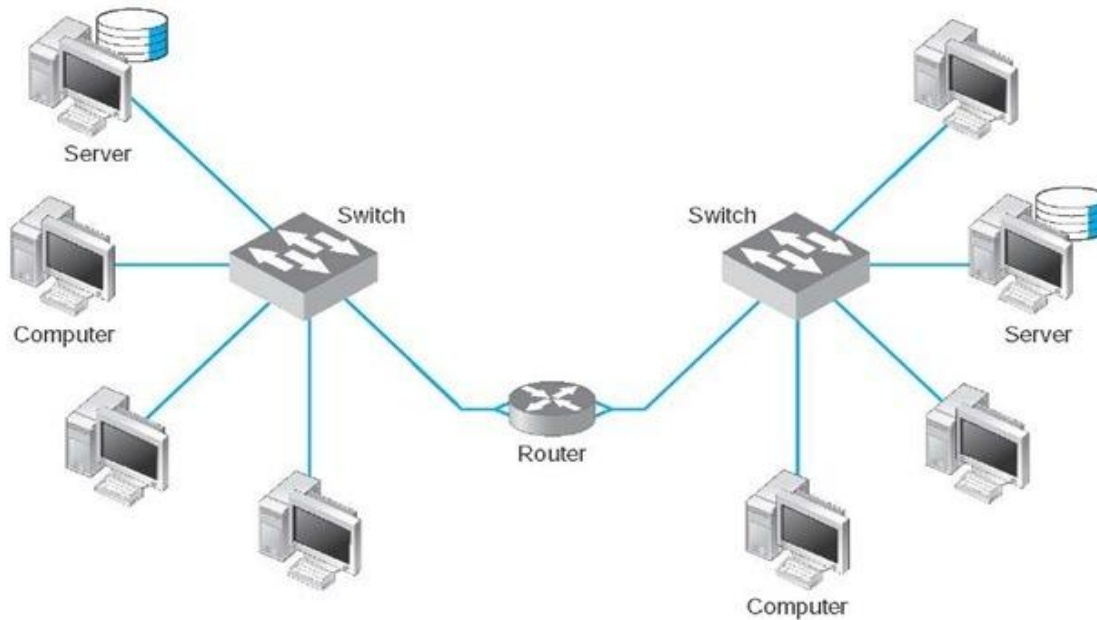


Figure 3

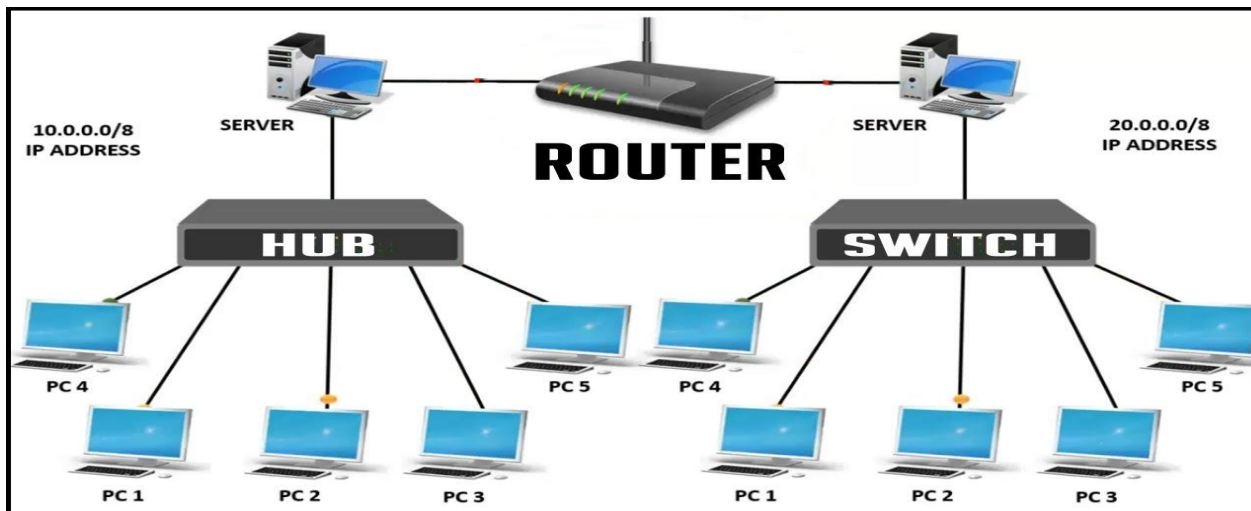
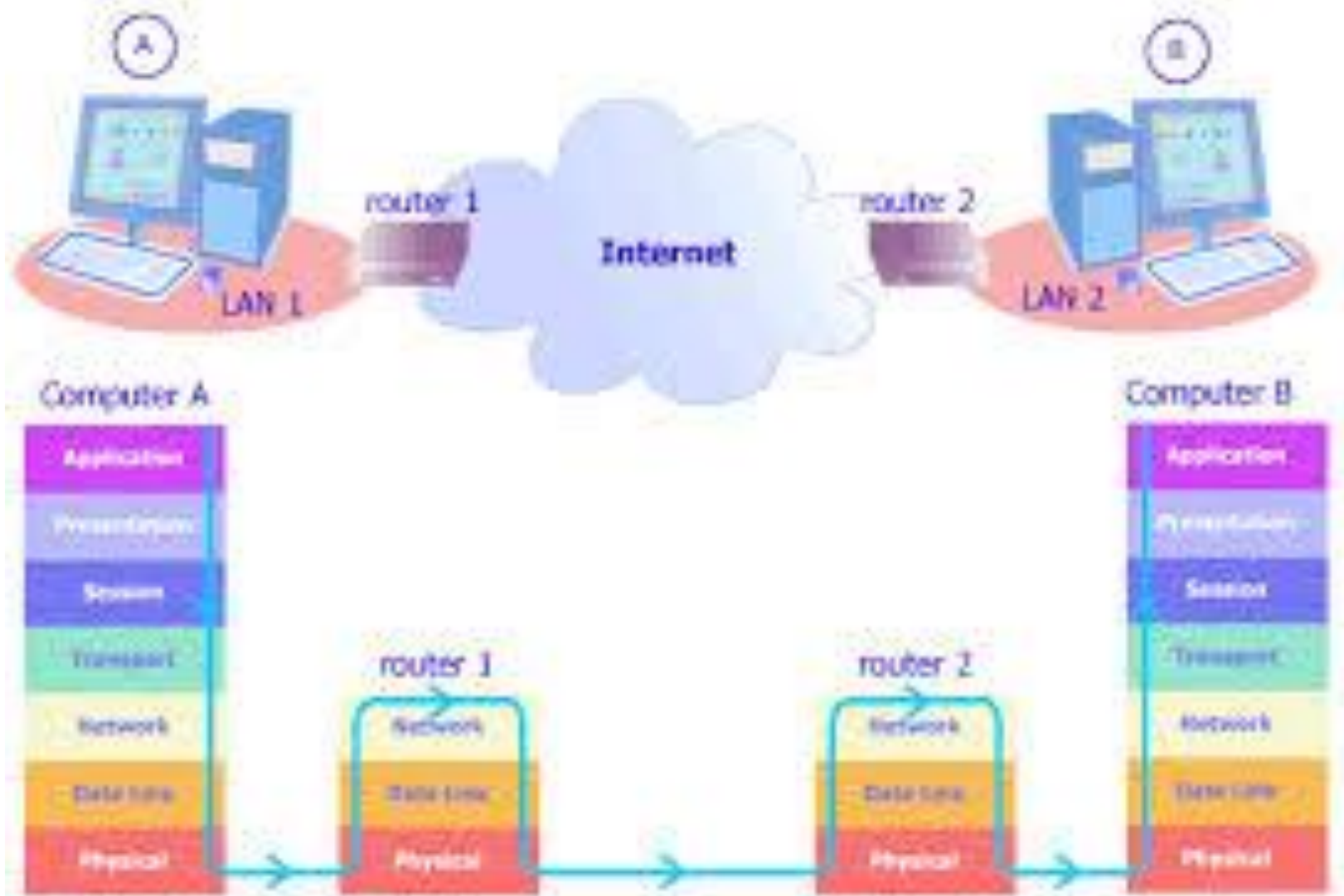


Figure 4

❖ Router

- A router is hardware device designed to receive, analyze and move incoming **packets** to another network.
- It operates in **physical, data link and network** layer of the OSI model. But most **active in network layer**.
- Routers are able to access network layer address (IP address) of the device.
- Simple function of router is to receive the **packet** from one connected network and pass them to a second connected network.
- Routers also perform the **traffic directing functions** on the Internet.
- A packet sent from a station on one network to a device on a neighboring network **goes first to the jointly device – router** which forwards the packet to the destination network.
- Routers consult with **routing table** when packet is ready to be forwarded.

Routers in OSI model



❖ Router characteristics

- **Least-cost routing:** Router finds the shortest path for the packet which is fastest, cheapest, reliable and secure.
- **Non Adaptive routing:** In which, once a pathway to a destination has been selected, the router sends all packets for that destination along that one route.
- **Adaptive routing:** In this, Router may select a **new route for each packet**. Routers send the packet depending on which route is most efficient at that moment.
- **Example:**
 - Packet transmission from **A to D**
 - Non adaptive:
 - Once path A-B-D is selected then all packets will follow the same path.
 - Adaptive:
 - First through network B, means A-B-D
 - Second through network C, means A-C-D
 - Third through network E, means A-E-D

- Routers find the best path and route the packets based on the information **stored in routing table**.
- Routing table also contains → **network address** of each connected device, **possible paths** and **cost of sending data**.

❖ **Types of router (Two types)**

↳ **Static router**

- In it, routing table information is entered manually. Administrator enters the route for each destination into the table.
- In case of change in the connection, it cannot update automatically.
- It is more secure. It always uses the same route.

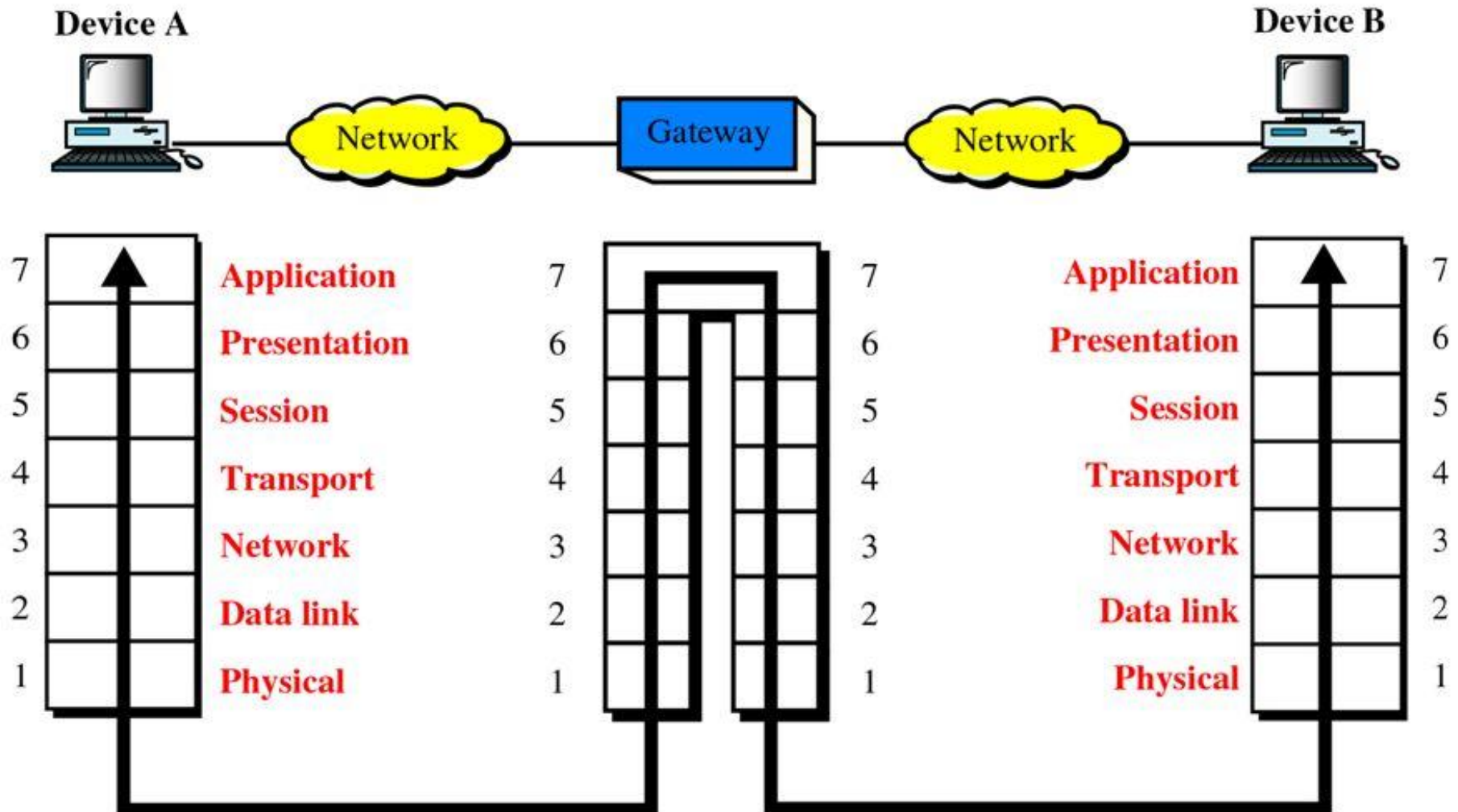
↳ **Dynamic router**

- In it, routing table is created and updated automatically.
- When there is a change in the connection, routing table is updated using any of the routing protocols.

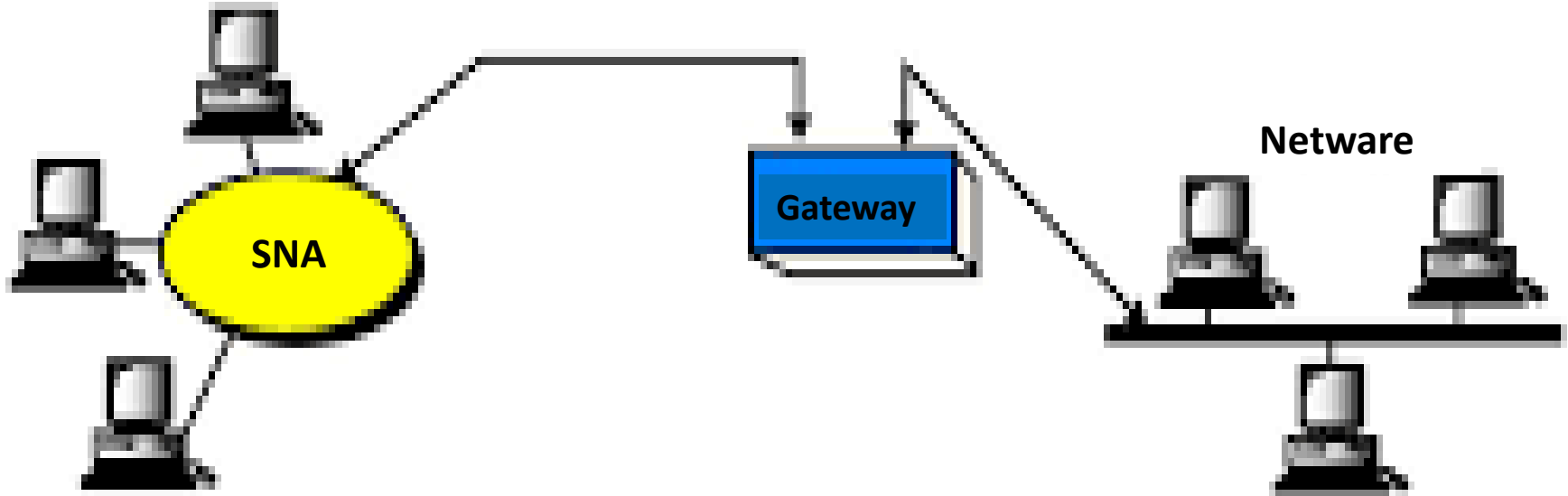
Gateways

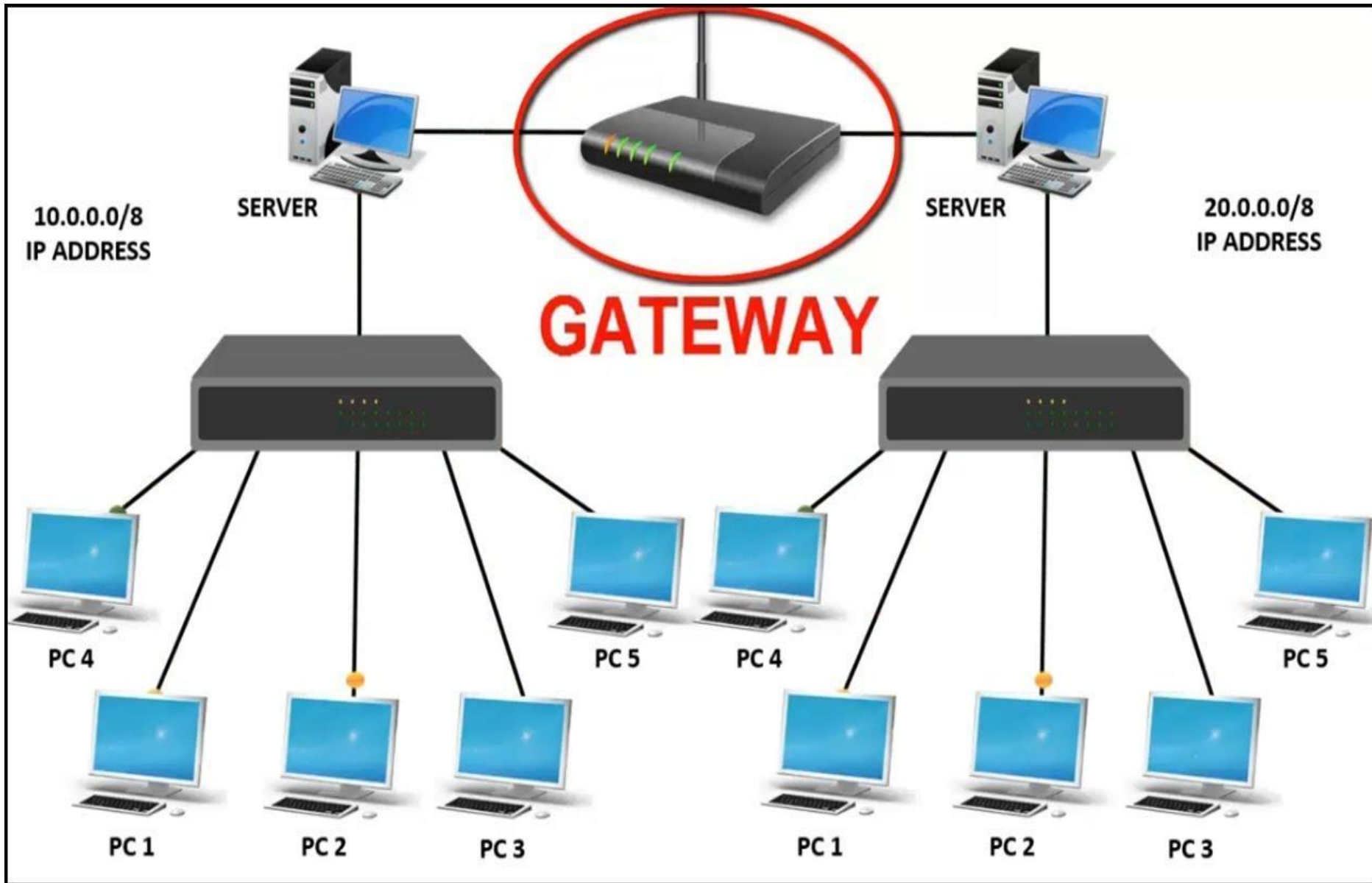
- Gateways operate in all seven layers of OSI model.
- A gateway is a hardware device that acts as a "gate" between two networks
- It is also called as **protocol convertor**.
- It is generally used to connect **two different network systems**.
- Routers transfers packets only across networks using **similar protocols** while a gateway can accept a packet formatted for one protocol and converts in to a packet formatted for another protocol before forwarding it.
- It may be a router, firewall, server, or other device that enables traffic to flow in and out of the network.

A Gateway in the OSI Model



Gateway as protocol convertor





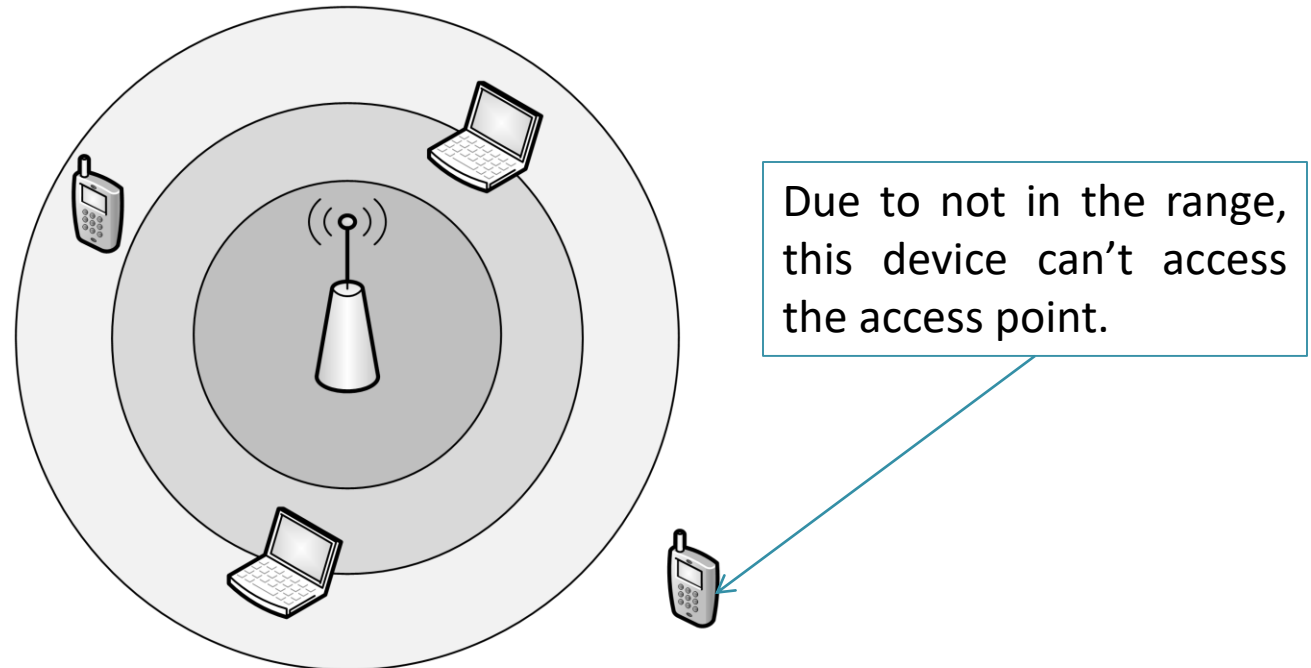
Access point

- **Definition:** An access point is a device that allows wireless devices to connect to a wired network using Wi-Fi.
- Most access points have built-in routers, while others must be connected to a router in order to provide network access.



- Access points can be found in many places, including houses, businesses, and public locations.

- Every access point has its own range. Any device placed within the range of access point can only use the services of that. (see below figure)



➡ How do access point works?

- Wireless access point performs the technique called '**modulation**' to transfer the data receives in or out.
- It takes the modulated radio frequency voltages highs and lows transmitted from a nearby wireless network card, and turns them into 1s and 0s instead. These 1s and 0s assembled into packets and forwarded to the destination.

Thank YOU...

Workstation (client or node)

- A workstation (WS) is a computer dedicated to a user or group of users for particular work. It includes one or more high resolution displays and a **faster processor than a personal computer (PC)**.
- In networking term → workstation is an individual single user computer which is connected with the server.

↪ There are two types of workstation:

1. *Diskless WS* → - a node which has no floppy drive, no HDD is referred as diskless node.
 - it has add-on card added with CPU and also has one boot RAM for temporary data storage.
2. *WS with HDD* → - programs can be loaded in hard disk. It has memory for permanent data storage.

Concept of wireless network

- Wireless network Is Collection of NICs, APs (access points) and routers.

↪ **Advantages of wireless network**

- Saves the cost of installation of LAN cabling.
- Easy to relocate the network.
- Easy to modify the structure of network.
- Easy to expand.

↪ **Disadvantages of wireless network**

- Troubleshooting is difficult.
- More expensive.

↪ **Types of wireless network**

- LAN
- Mobile computing

Types of communication

1. Unicasting

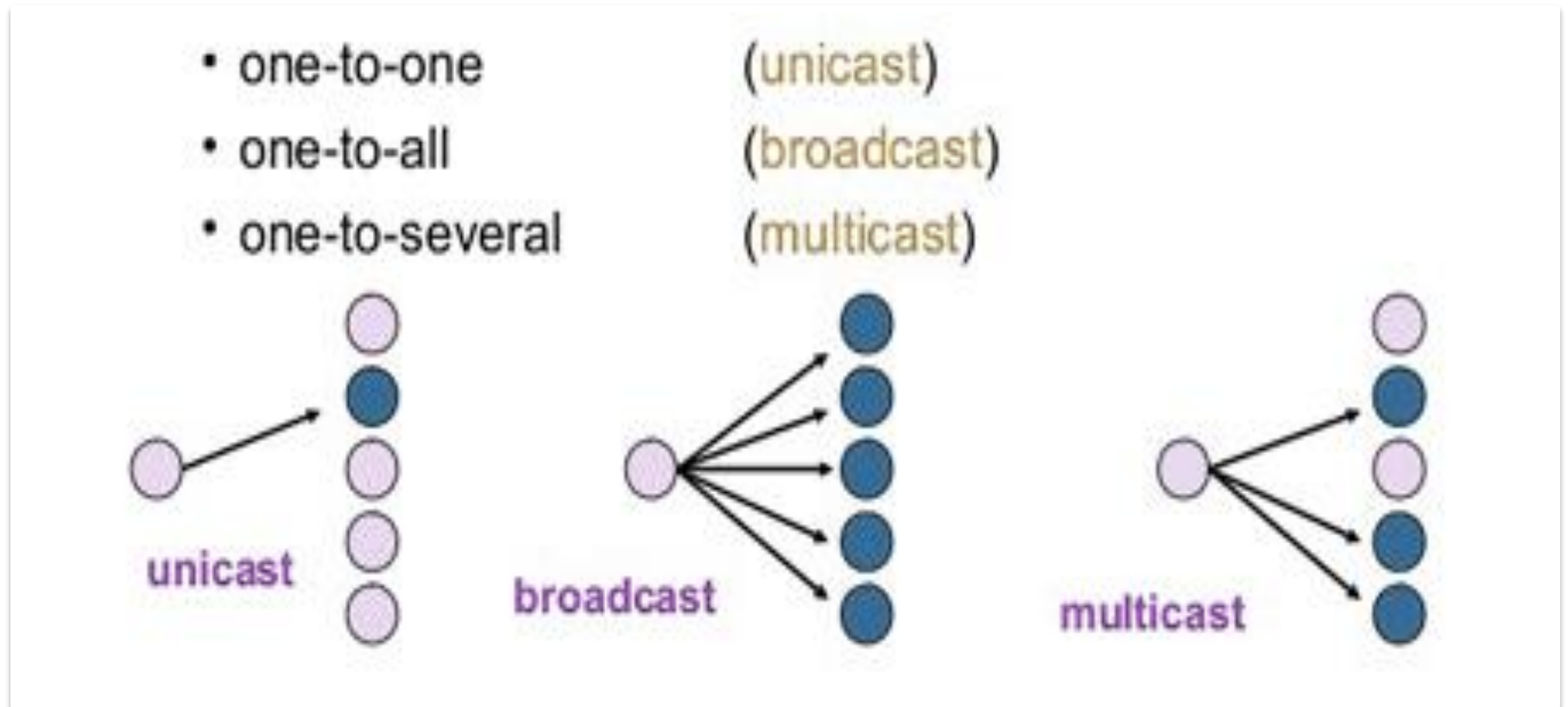
- Unicast is communication between a single sender and a single receiver over a network.
- In other term it can be said '*point-to-point*' communication.
- Examples of unicasting are: HTTP, TELNET, POP3, SSH

2. Multicasting

- Multicast is communication between a single sender and a a group of users over a network.
- Examples of multicasting are: video conferencing, e-mail system, teleconferencing.

3. Broadcasting

- In broadcast, data is delivered from one point to several points.
- Communication between sender to all the nodes (including itself).
- Example → satellite television, education video broadcasting.

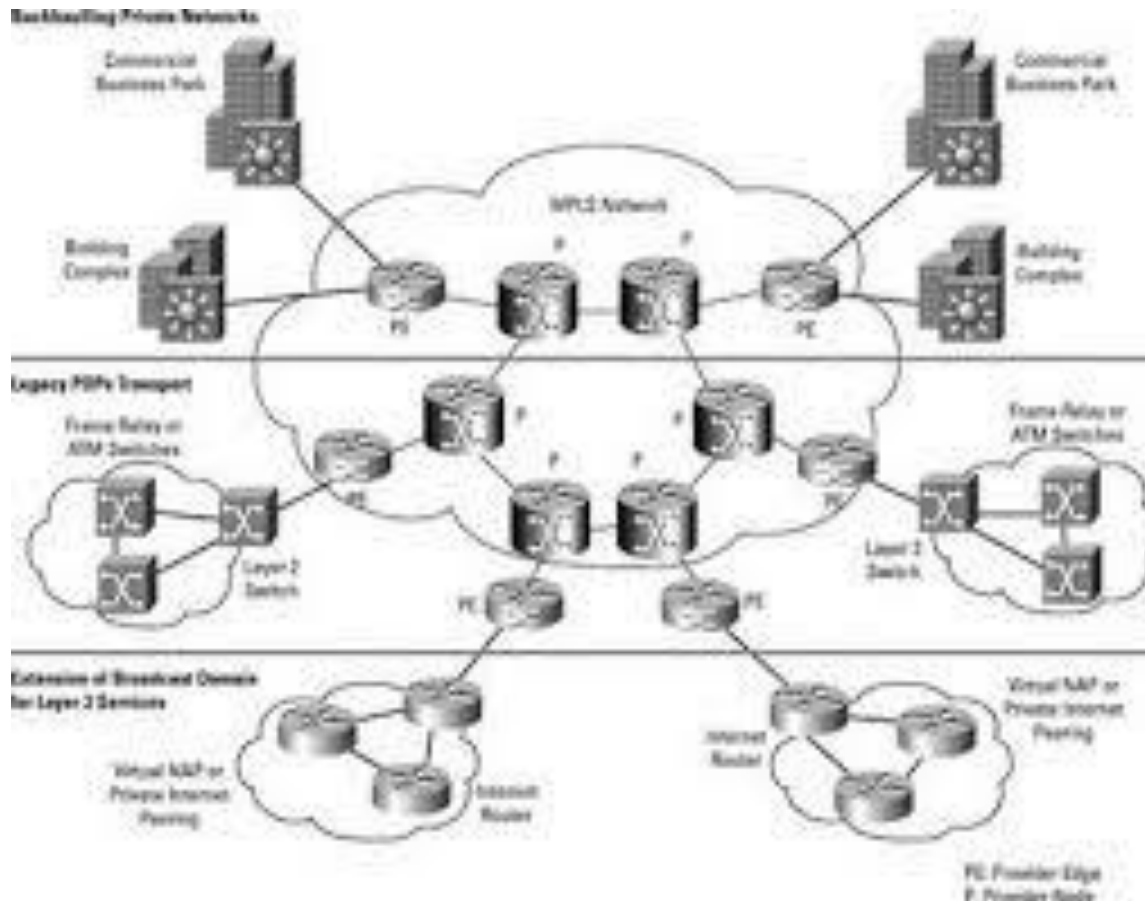


LEARNING OUTCOME (LO)

- After completion of this session, students will be able to *explain network management software.*

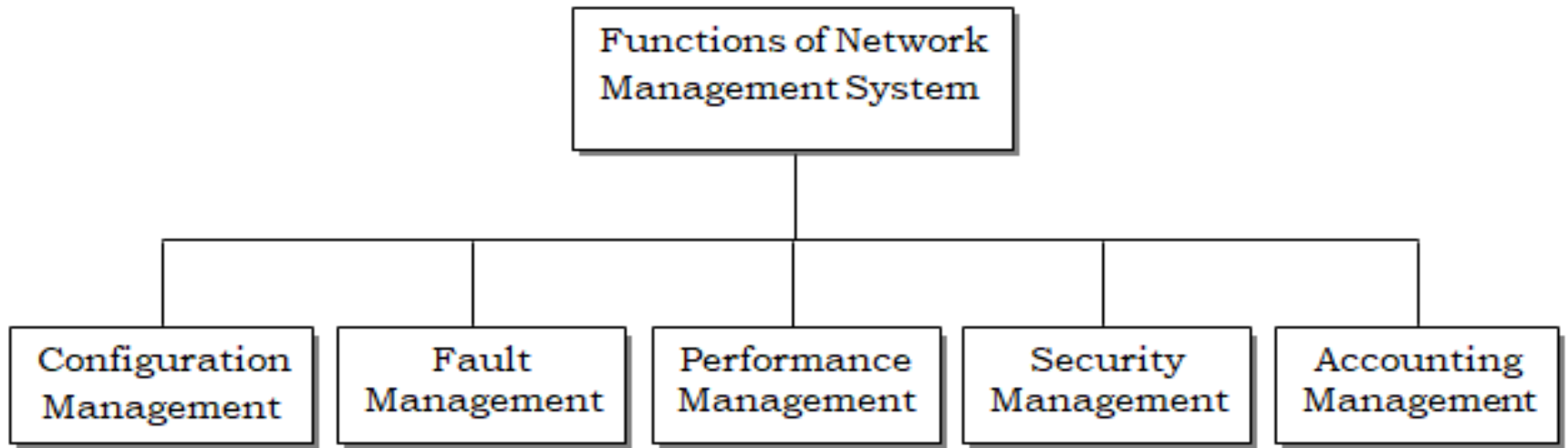
Network Management Software

- When network becomes more complex in terms of device population, topology and distances, there should be some way for controlling all the pieces of whole network.



- **Network management** can be defined as monitoring, testing, configuring, and troubleshooting network components to meet a set of requirements defined by an organization.
- Two standard protocols are there for network management and monitoring.
 1. **SNMP(Simple Network Management Protocol)**
 - It defines the set of instructions that can be used to view and change the settings of any SNMP compliant device.
 2. **RMON (Remote MONitoring protocol)**
 - It allows remote collection of network information.

❖ Basic features (functions) of network management software



➡ Configuration management

- In which network operation is monitored and controlled.
- Addition of new programs and equipments, modification of existing systems are coordinated. And all these are updated regularly.

↪ **Fault management**

- In it, network problems are found and corrected.
- Future problems are identified and prevented from re occurring.
- Network downtime is minimized and keep network in operational mode most time.

↪ **Performance management**

- In it, the capacity, traffic, throughput, or response time of the network is measured and improve the performance of the network.

↪ **Security management**

- In it, network is protected against hackers and unauthorized users for physical/ electronic damage.
- User confidentiality is maintained.

↪ **Accounting management**

- In which, distributing resources optimally and fairly among network subscribers.
- Making most effective use of available systems.
- Minimizing cost of operations.